

Tencent HY

API Documentation

Product Documentation



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Modified APIs:

- [SubmitHunyuanTo3DRapidJob](#)
 - New input parameters:ResultFormat

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- [DescribeHunyuanTo3DUVJob](#)
- [SubmitHunyuanTo3DUVJob](#)

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- [QueryHunyuanTo3DTextureEditJob](#)

- [SubmitHunyuanTo3DTextureEditJob](#)

New data structures:

- [ImageInfo](#)

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New data structures:

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- [PromptTokensDetails](#)
- [Reference](#)
- [TranslationChoice](#)
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- [Usage](#)

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- [Describe3DSmartTopologyJob](#)
- [Submit3DSmartTopologyJob](#)

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- [SubmitHunyuanTo3DProJob](#)
 - New input parameters:Model

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New APIs:

- [Convert3DFormat](#)

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- [QueryHunyuan3DPartJob](#)
- [QueryHunyuanTo3DRapidJob](#)
- [SubmitHunyuan3DPartJob](#)
- [SubmitHunyuanTo3DRapidJob](#)

New data structures:

- [InputFile3D](#)

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New APIs:

- [QueryHunyuanTo3DProJob](#)
- [SubmitHunyuanTo3DProJob](#)

New data structures:

- [File3D](#)
- [ViewImage](#)

Introduction

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Overview

Tencent Hunyuan is fully self-developed by Tencent, delivering outstanding performance in high-quality content creation, mathematical logic, code generation, and multi-round dialogue, achieving industry-leading standards. The Hunyuan API supports AI search with online plug-ins, integrating Tencent's high-quality content ecosystem to offer powerful real-time, in-depth content retrieval and AI FAQ capability.

For all supported API information of Tencent Hunyuan, please refer to the [API overview](#).

Glossary

Term	Description
Concurrent task count refers to the number of tasks that can be processed simultaneously. If the concurrent task count is 1, only 1 submitted task can be processed simultaneously, and the next task can be processed only after the previous task is completed.	
API key. The API key is a secure credential required for user access to Tencent Cloud API to perform identity verification, consisting of SecretId and SecretKey.	

Usage limits

For API parameter limits, refer to the parameter descriptions in each API document.

Getting Started with APIs

You can use the API Explorer tool to call APIs online.

This document uses application creation as an example. The steps to make an API call via the API Explorer Tool are as follows:

1. Go to the [API Explorer](#) page. For more API Explorer Tool usage information, see the [document](#).
2. Call the [ChatCompletions](#) API to invoke Tencent Hunyuan.

API Category

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Hunyuan Text APIs

API Name	Feature	Frequency Limit (maximum requests per second)
ChatTranslations	Translate	20

Hunyuan 3D APIs

API Name	Feature	Frequency Limit (maximum requests per second)
SubmitHunyuanTo3DProJob	Submit hunyuan 3D pro edition task	20
QueryHunyuanTo3DProJob	Query a hunyuan 3D pro edition task	20
SubmitHunyuan3DPartJob	Submit component generation task	20
QueryHunyuan3DPartJob	Queries a component generation task	20
Submit3DSmartTopologyJob	Submit an intelligent topology task	20
Describe3DSmartTopologyJob	Query intelligent topology tasks	20
SubmitHunyuanTo3DRapidJob	Submit hunyuan 3D ultra-fast version task	20
QueryHunyuanTo3DRapidJob	Query a hunyuan 3D ultra-fast version task	20
SubmitHunyuanTo3DTextureEditJob	Submit a texture edit task	20
QueryHunyuanTo3DTextureEditJob	Query texture edit tasks	20

SubmitHunyuanTo3DUVJob	Submit a UV unfold task	20
DescribeHunyuanTo3DUVJob	Query a UV unfold task	20
Convert3DFormat	Switch the model format	20

Making API Requests

Request Structure

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1. Service Address

The API supports access from either a nearby region (at `hunyuan.intl.tencentcloudapi.com`) or a specified region (at `hunyuan.ap-guangzhou.tencentcloudapi.com` for Guangzhou, for example).

We recommend using the domain name to access the nearest server. When you call an API, the request is automatically resolved to a server in the region **nearest** to the location where the API is initiated. For example, when you initiate an API request in Guangzhou, this domain name is automatically resolved to a Guangzhou server, the result is the same as that of specifying the region in the domain like "`hunyuan.ap-guangzhou.tencentcloudapi.com`".

Note: For latency-sensitive businesses, we recommend that you specify the region in the domain name.

Tencent Cloud currently supports the following regions:

Hosted region	Domain name
Local access region (recommended, only for non-financial availability zones)	<code>hunyuan.intl.tencentcloudapi.com</code>
South China (Guangzhou)	<code>hunyuan.ap-guangzhou.tencentcloudapi.com</code>
East China (Shanghai)	<code>hunyuan.ap-shanghai.tencentcloudapi.com</code>
East China (Nanjing)	<code>hunyuan.ap-nanjing.tencentcloudapi.com</code>
North China (Beijing)	<code>hunyuan.ap-beijing.tencentcloudapi.com</code>
Southwest China (Chengdu)	<code>hunyuan.ap-chengdu.tencentcloudapi.com</code>
Southwest China (Chongqing)	<code>hunyuan.ap-chongqing.tencentcloudapi.com</code>
Hong Kong, Macao, Taiwan (Hong Kong, China)	<code>hunyuan.ap-hongkong.tencentcloudapi.com</code>

Southeast Asia (Singapore)	hunyuan.ap-singapore.tencentcloudapi.com
Southeast Asia (Jakarta)	hunyuan.ap-jakarta.tencentcloudapi.com
Southeast Asia (Bangkok)	hunyuan.ap-bangkok.tencentcloudapi.com
Northeast Asia (Seoul)	hunyuan.ap-seoul.tencentcloudapi.com
Northeast Asia (Tokyo)	hunyuan.ap-tokyo.tencentcloudapi.com
U.S. East Coast (Virginia)	hunyuan.na-ashburn.tencentcloudapi.com
U.S. West Coast (Silicon Valley)	hunyuan.na-siliconvalley.tencentcloudapi.com
South America (São Paulo)	hunyuan.sa-saopaulo.tencentcloudapi.com
Europe (Frankfurt)	hunyuan.eu-frankfurt.tencentcloudapi.com

2. Communications Protocol

All the Tencent Cloud APIs communicate via HTTPS, providing highly secure communication tunnels.

3. Request Methods

Supported HTTP request methods:

- POST (recommended)
- GET

The Content-Type types supported by POST requests:

- application/json (recommended). The TC3-HMAC-SHA256 signature algorithm must be used.
- application/x-www-form-urlencoded. The HmacSHA1 or HmacSHA256 signature algorithm must be used.
- multipart/form-data (only supported by certain APIs). You must use TC3-HMAC-SHA256 to calculate the signature.

The size of a GET request packet is up to 32 KB. The size of a POST request is up to 1 MB when the HmacSHA1 or HmacSHA256 signature algorithm is used, and up to 10 MB when TC3-HMAC-SHA256 is used.

4. Character Encoding

Only UTF-8 encoding is used.

Common Params

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Common parameters are used for all APIs authenticating requestors. Common parameters must be included in all API requests, and they will not be described in individual API documents.

The exact contents of the common parameters will vary depending on the version of the signature method you use.

Common parameters for Signature Algorithm v3

When the TC3-HMAC-SHA256 algorithm is used, the common parameters should be uniformly placed in the HTTP request header, as shown below:

Parameter Name	Type	Required	Description
X-TC-Action	String	Yes	The name of the API for the desired operation. For the specific value, see the description of common parameter <code>Action</code> in the input parameter documentation. For example, the API for querying the CVM instances is <code>DescribeInstances</code> .
X-TC-Region	String	Yes	Region parameter, which is used to identify the region to which the request work with belongs. For values supported for an API, see the description of parameter <code>Region</code> in the input parameters in related API documentation. This parameter is not required for some APIs (which will be indicated in the documentation), and will not take effect even it is passed.
X-TC-Timestamp	Integer	Yes	The current UNIX timestamp that records the time when the API request is sent, for example, 1529223702. Note: If the difference between the UNIX timestamp and server time is greater than 5 minutes, a signature expiration error will be returned.
X-TC-Version	String	Yes	API version of the action. For the valid values, see the description of input parameter <code>Version</code> in the API documentation. For example, the CVM API version is 2017-03-12.
Authorization	String	Yes	The HTTP authentication request header, for example: TC3-HMAC-SHA256 Credential=AKID***/Date/service/tc3_request_headers;host, Signature=fe5f80f77d5fa3beca038a248ff027d0445342fe2855ddc Here: - TC3-HMAC-SHA256: Signature method, currently fixed as this

			– Credential: Signature credential; AKID*** is the SecretId; Date is and this value must match the value of X-TC-Timestamp (a common UTC time format; service is the name of the product/service, and domain name prefix. For example, a domain name cvm.tencentcloudapi.com the CVM product and the value would be cvm; – SignedHeaders: The headers that contains the authentication information type and host are the required headers; – Signature: Signature digest.
X-TC-Token	String	No	The token used for a temporary certificate. It must be used with a key to obtain the temporary key and token by calling a CAM API. No long-term key.

Assuming you want to query the list of Cloud Virtual Machine instances in the Guangzhou region, the request structure in the form of request URL, request header and request body may be as follows:

Example of an HTTP GET request structure:

```
https://cvm.tencentcloudapi.com/?Limit=10&Offset=0

Authorization: TC3-HMAC-SHA256 Credential=AKID*****/2018-10-09/cvm/tc3_request, SignedHeaders=content-type;host, Signature=5da7a33f6993f0614b047e5df4582db9e9bf4672ba50567dba16c6ccf174c474
Content-Type: application/x-www-form-urlencoded
Host: cvm.tencentcloudapi.com
X-TC-Action: DescribeInstances
X-TC-Version: 2017-03-12
X-TC-Timestamp: 1539084154
X-TC-Region: ap-guangzhou
```

The following example shows you how to structure an HTTP POST (application/json) request:

```
https://cvm.tencentcloudapi.com/

Authorization: TC3-HMAC-SHA256 Credential=AKID*****/2018-05-30/cvm/tc3_request, SignedHeaders=content-type;host, Signature=582c400e06b5924a6f2b5d7d672d79c15b13162d9279b0855cfba6789a8edb4c
Content-Type: application/json
Host: cvm.tencentcloudapi.com
X-TC-Action: DescribeInstances
X-TC-Version: 2017-03-12
X-TC-Timestamp: 1527672334
X-TC-Region: ap-guangzhou

{"Offset":0,"Limit":10}
```

Example of an HTTP POST (multipart/form-data) request structure (only supported by specific APIs):

```
https://cvm.tencentcloudapi.com/

Authorization: TC3-HMAC-SHA256 Credential=AKID*****/20
18-05-30/cvm/tc3_request, SignedHeaders=content-type;host, Signature=582c400e06b5
924a6f2b5d7d672d79c15b13162d9279b0855cfba6789a8edb4c
Content-Type: multipart/form-data; boundary=58731222010402
Host: cvm.tencentcloudapi.com
X-TC-Action: DescribeInstances
X-TC-Version: 2017-03-12
X-TC-Timestamp: 1527672334
X-TC-Region: ap-guangzhou

--58731222010402
Content-Disposition: form-data; name="Offset"

0
--58731222010402
Content-Disposition: form-data; name="Limit"

10
--58731222010402--
```

Common parameters for Signature Algorithm v1

To adopt the HmacSHA1 and HmacSHA256 signature methods, common parameters must be put into the request string, as shown below:

Parameter Name	Type	Required	Description
Action	String	Yes	The name of the API for the desired operation. For the specific value, see the description of common parameter <code>Action</code> in the input parameters in related API documentation. For example, the API for querying the CVM instance list is <code>DescribeInstances</code> .
Region	String	Yes	Region parameter, which is used to identify the region to which the data you want to work with belongs. For values supported for an API, see the description of common parameter <code>Region</code> in the input parameters in related API documentation. Note: This parameter is not required for

			some APIs (which will be indicated in related API documentation), and will not take effect even if it is passed.
Timestamp	Integer	Yes	The current UNIX timestamp that records the time when the API request was initiated, for example, 1529223702. If the difference between the value and the current system time is too large, a signature expiration error may occur.
Nonce	Integer	Yes	A random positive integer used along with <code>Timestamp</code> to prevent replay attacks.
SecretId	String	Yes	The identifying SecretId obtained on the Cloud API Key page. A SecretId corresponds to a unique SecretKey which is used to generate the request signature (Signature).
Signature	String	Yes	Request signature used to verify the validity of this request. This is calculated based on the actual input parameters. For more information about how this is calculated, see the API authentication documentation.
Version	String	Yes	API version of the action. For the valid values, see the description of the common input parameter <code>Version</code> in the API documentation. For example, the version of CVM is 2017-03-12.
SignatureMethod	String	No	Signature method. Currently, only HmacSHA256 and HmacSHA1 are supported. The HmacSHA256 algorithm is used to verify the signature only when this parameter is specified as HmacSHA256. In other cases, the signature is verified with HmacSHA1.
Token	String	No	The token used for a temporary certificate. It must be used with a temporary key. You can obtain the temporary key and token by calling a CAM API. No token is required for a long-term key.

Assuming you want to query the list of Cloud Virtual Machine instances in the Guangzhou region, the request structure in the form of request URL, request header and request body may be as follows:

Example of an HTTP GET request structure:

```
https://cvm.tencentcloudapi.com/?Action=DescribeInstances&Version=2017-03-12&SignatureMethod=HmacSHA256&Timestamp=1527672334&Signature=37ac2f4fde00b0ac9bd9eadeb459b1bbec224158d66e7ae5fcadb70b2d181d02&Region=ap-guangzhou&Nonce=23823223&SecretId=AKID*****
```

```
Host: cvm.tencentcloudapi.com
Content-Type: application/x-www-form-urlencoded
```

Example of an HTTP POST request structure:

```
https://cvm.tencentcloudapi.com/

Host: cvm.tencentcloudapi.com
Content-Type: application/x-www-form-urlencoded

Action=DescribeInstances&Version=2017-03-12&SignatureMethod=HmacSHA256&Timestamp=
1527672334&Signature=37ac2f4fde00b0ac9bd9eadeb459b1bbbee224158d66e7ae5fcadb70b2d18
1d02&Region=ap-guangzhou&Nonce=23823223&SecretId=AKID*****
****
```

Region List

The supported Region field values for all APIs in this product are listed as below. For any API that does not support any of the following regions, this field will be described additionally in the relevant API document.

Region	Value
Southeast Asia (Singapore)	ap-singapore

Signature v3

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TencentCloud API authenticates every single request, i.e., the request must be signed using the security credentials in the designated steps. Each request has to contain the signature information (Signature) in the common request parameters and be sent in the specified way and format.

Applying for Security Credentials

The security credential used in this document is a key, which includes a SecretId and a SecretKey. Each user can have up to two pairs of keys.

- SecretId: Used to identify the API caller, which is just like a username.
- SecretKey: Used to authenticate the API caller, which is just like a password.
- **You must keep your security credentials private and avoid disclosure; otherwise, your assets may be compromised. If they are disclosed, please disable them as soon as possible.**

You can apply for the security credentials through the following steps:

1. Log in to the [Tencent Cloud Console](#).
2. Go to the [TencentCloud API Key](#) console page.
3. On the [TencentCloud API Key](#) page, click **Create** to create a SecretId/SecretKey pair.

Using the Resources for Developers

TencentCloud API comes with SDKs for seven commonly used programming languages, including [Python](#), [Java](#), [PHP](#), [Go](#), [NodeJS](#) and [.NET](#). In addition, it provides [API Explorer](#) which enables online call, signature verification, and SDK code generation. If you have any troubles calculating a signature, consult these resources.

TC3-HMAC-SHA256 Signature Algorithm

Compatible with the previous HmacSHA1 and HmacSHA256 signature algorithms, the TC3-HMAC-SHA256 signature algorithm is more secure and supports larger requests and JSON format with better performance. We recommend using TC3-HMAC-SHA256 to calculate the signature.

TencentCloud API supports both GET and POST requests. For the GET method, only the Content-Type: application/x-www-form-urlencoded protocol format is supported. For the POST method, two protocol formats, Content-Type: application/json and Content-Type: multipart/form-data, are supported. The JSON format is supported by default for all business APIs, and the multipart format is supported only for specific business APIs. In this case, the API cannot be called in JSON format. See the specific business API documentation for more information. The POST method is recommended, as there is no difference in the results of both the methods, but the GET method only supports request packets up to 32 KB.

The following uses querying the list of CVM instances in the Guangzhou region as an example to describe the steps of signature splicing. We chose this API because:

1. CVM is activated by default, and this API is often used;
2. It is read-only and does not change the status of existing resources;
3. It covers many types of parameters, which allows it to be used to demonstrate how to use arrays containing data structures.

In the example, we try to choose common parameters and API parameters that are prone to mistakes. When you actually call an API, please use parameters based on the actual conditions. The parameters vary by API. Do not copy the parameters and values in this example.

Assuming that your SecretId and SecretKey are `AKID*****` and `*****`, respectively, if you want to view the status of the instance in the Guangzhou region whose CVM instance name is "unnamed" and have only one data entry returned, then the request may be:

```
curl -X POST https://cvm.tencentcloudapi.com \
-H "Authorization: TC3-HMAC-SHA256 Credential=AKID*****/2019-02-25/cvm/tc3_request, SignedHeaders=content-type;host, Signature=a7b8551448762bd123d6f79e81815e31a92013640a6cef36a08ad4b292a4d2f2" \
-H "Content-Type: application/json; charset=utf-8" \
-H "Host: cvm.tencentcloudapi.com" \
-H "X-TC-Action: DescribeInstances" \
-H "X-TC-Timestamp: 1551113065" \
-H "X-TC-Version: 2017-03-12" \
-H "X-TC-Region: ap-guangzhou" \
-d '{"Limit": 1, "Filters": [{"Values": ["unnamed"], "Name": "instance-name"}]}'
```

The signature calculation process is explained in detail below.

1. Concatenating the CanonicalRequest String

Concatenate the canonical request string (CanonicalRequest) in the following pseudocode format:

```
CanonicalRequest =
HTTPRequestMethod + '\n' +
CanonicalURI + '\n' +
CanonicalQueryString + '\n' +
CanonicalHeaders + '\n' +
SignedHeaders + '\n' +
HashedRequestPayload
```

Field Name	Explanation
HTTPRequestMethod	HTTP request method (GET or POST). This example uses <code>POST</code> .
CanonicalURI	URI parameter. Slash ("/") is used for API 3.0.
CanonicalQueryString	The query string in the URL of the originating HTTP request. This is always an empty string "" for POST requests, and is the string after the question mark (?) for GET requests. For example: <code>Limit=10&Offset=0</code>. Note: <code>CanonicalQueryString</code> must be URL-encoded, referencing RFC3986 and the UTF8 character set. We recommend using the programming language library. All special characters must be encoded and capitalized.
CanonicalHeaders	Header information for signature calculation, including at least two headers of <code>host</code> and <code>content-type</code>. Custom headers can be added to participate in the signature process to improve the uniqueness and security of the request. Concatenation rules: - Both the key and value of the header should be converted to lowercase with leading and trailing spaces removed, so they are concatenated in the format <code>key:value\n</code> format; - If there are multiple headers, they should be sorted in ASCII ascending order of the header keys (lowercase). The calculation result in this example is <code>content-type:application/json; charset=utf-8\nhost:cvm.tencentcloudapi.com\n</code>. Note: <code>content-type</code> must match the actually sent content. In some programming languages, a charset value would be added even if it is not specified. In this case, the request sent is different from the one signed, and the server will return an error indicating that signature verification failed.
SignedHeaders	Header information for signature calculation, indicating which headers of the request participate in the signature process (they must each individually correspond to the headers in CanonicalHeaders). <code>Content-type</code> and <code>host</code> are required headers. Concatenation rules: - Both the key and value of the header should be converted to lowercase; - If there are multiple headers, they should be sorted in ASCII ascending order of the header keys (lowercase) and separated by semicolons (;).

	The value in this example is <code>content-type;host</code>
HashedRequestPayload	Hash value of the request payload (i.e., the body, such as <code>{"Limit": 1, "Filters": [{"Values": ["unnamed"], "Name": "instance-name"}]}</code> this example). The pseudocode for calculation is <code>Lowercase(HexEncode(Hash.SHA256(RequestPayload)))</code> by SHA256 hashing the payload of the HTTP request, performing hexadecimal encoding, and finally converting the encoded string to lowercase letters. For GET requests, <code>RequestPayload</code> is always an empty string. The calculation result in this example is <code>99d58dfbc6745f6747f36bfca17dee5e6881dc0428a0a36f96199342bc5b4</code>

According to the rules above, the `CanonicalRequest` string obtained in the example is as follows:

POST

/

content-type:application/json; charset=utf-8

host:cvm.tencentcloudapi.com

content-type;host

99d58dfbc6745f6747f36bfca17dee5e6881dc0428a0a36f96199342bc5b4907

2. Concatenating the String to Be Signed

The string to sign is concatenated as follows:

```
StringToSign =
Algorithm + \n +
RequestTimestamp + \n +
CredentialScope + \n +
HashedCanonicalRequest
```

Field Name	Explanation
Algorithm	Signature algorithm, which is currently always <code>TC3-HMAC-SHA256</code> .
RequestTimestamp	Request timestamp, i.e., the value of the common parameter <code>X-TC-Timestamp</code> in the request header, which is the UNIX timestamp of the current time in seconds, such as <code>1551113065</code> in this example.
CredentialScope	Scope of the credential in the format of <code>Date/service/tc3_request</code> , including the date, requested service and termination string (tc3_request). Date is a UTC time, whose value should match the UTC date converted by the common parameter <code>X-TC-Timestamp</code> ; <code>service</code> is the product name, which should

	match the domain name of the product called. The calculation result in this example is <code>2019-02-25/cvm/tc3_request</code> .
HashedCanonicalRequest	Hash value of the CanonicalRequest string concatenated in the steps above. The pseudocode for calculation is <code>Lowercase(HexEncode(Hash.SHA256(CanonicalRequest)))</code> . The calculation result in this example is <code>2815843035062fffd6f2a44ea8a34818b0dc46f024b8b3786976a3adda7a</code> .

Note:

1. Date has to be calculated from the timestamp "X-TC-Timestamp" and the time zone is UTC+0. If you add the system's local time zone information (such as UTC+8), calls can succeed both day and night but will definitely fail at 00:00. For example, if the timestamp is 1551113065 and the time in UTC+8 is 2019-02-26 00:44:25, the UTC+0 date in the calculated Date value should be 2019-02-25 instead of 2019-02-26.
2. Timestamp must be the same as your current system time, and your system time and standard time must be synced; if the difference between Timestamp and your current system time is larger than five minutes, the request will fail. If your system time is out of sync with the standard time for a while, the request will fail and return a signature expiration error.

According to the preceding rules, the string to be signed obtained in the example is as follows:

```
TC3-HMAC-SHA256
1551113065
2019-02-25/cvm/tc3_request
2815843035062fffd6f2a44ea8a34818b0dc46f024b8b3786976a3adda7a
```

3. Calculating the Signature

1) Calculate the derived signature key with the following pseudocode:

```
SecretKey = "*****"
SecretDate = HMAC_SHA256("TC3" + SecretKey, Date)
SecretService = HMAC_SHA256(SecretDate, Service)
SecretSigning = HMAC_SHA256(SecretService, "tc3_request")
```

Field Name	Explanation
------------	-------------

SecretKey	The original SecretKey, i.e., ***** .
Date	The Date field information in Credential , such as 2019-02-25 in this example.
Service	Value in the Service field in Credential , such as cvm in this example.

2) Calculate the signature with the following pseudocode:

```
Signature = HexEncode(HMAC_SHA256(SecretSigning, StringToSign))
```

4. Concatenating the Authorization

The Authorization is concatenated as follows:

```
Authorization =
Algorithm + ' ' +
'Credential=' + SecretId + '/' + CredentialScope + ', ' +
'SignedHeaders=' + SignedHeaders + ', ' +
'Signature=' + Signature
```

Field Name	Explanation
Algorithm	Signature algorithm, which is always TC3-HMAC-SHA256 .
SecretId	The SecretId in the key pair, i.e., AKID***** .
CredentialScope	Credential scope (see above). The calculation result in this example is 2019-02-25/cvm/tc3_request .
SignedHeaders	Header information for signature calculation (see above), such as content-type;host in this example.
Signature	Signature value. The calculation result in this example is a7b8551448762bd123d6f79e81815e31a92013640a6cef36a08ad4b292a4d2f2 .

According to the rules above, the value obtained in the example is:

```
TC3-HMAC-SHA256 Credential=AKID*****/2019-02-25/cvm/tc3_request, SignedHeaders=content-type;host, Signature=a7b8551448762bd123d6f79e81815e31a92013640a6cef36a08ad4b292a4d2f2
```

The following example shows a finished authorization header:

```
POST https://cvm.tencentcloudapi.com/
Authorization: TC3-HMAC-SHA256 Credential=AKID*****/20
19-02-25/cvm/tc3_request, SignedHeaders=content-type;host, Signature=a7b855144876
2bd123d6f79e81815e31a92013640a6cef36a08ad4b292a4d2f2
Content-Type: application/json; charset=utf-8
Host: cvm.tencentcloudapi.com
X-TC-Action: DescribeInstances
X-TC-Version: 2017-03-12
X-TC-Timestamp: 1551113065
X-TC-Region: ap-guangzhou

{"Limit": 1, "Filters": [{"Values": ["unnamed"], "Name": "instance-name"}]}
```

5. Signature Demo

When calling API 3.0, you are recommended to use the corresponding Tencent Cloud SDK 3.0 which encapsulates the signature process, enabling you to focus on only the specific APIs provided by the product when developing. See [SDK Center](#) for more information. Currently, the following programming languages are supported:

- [Python](#)
- [Java](#)
- [PHP](#)
- [Go](#)
- [NodeJS](#)
- [.NET](#)

To further explain the signing process, we will use a programming language to implement the process described above. The request domain name, API and parameter values in the sample are used here. This goal of this example is only to provide additional clarification for the signature process, please see the SDK for actual usage.

The final output URL might be: `https://cvm.tencentcloudapi.com/?`

`Action=DescribeInstances&InstanceId=ins-09dx96dg&Limit=20&Nonce=11886&Offset=0&Region=ap-guangzhou&SecretId=AKID*****&Signature=ElIP9YW3pW28FpsEdkXt%2F%2BWcGel%3D&Timestamp=1465185768&Version=2017-03-12.`

Note: The key in the example is fictitious, and the timestamp is not the current time of the system, so if this URL is opened in the browser or called using commands such as curl, an authentication error will be returned: Signature expired. In order to get a URL that can work properly, you need to replace the SecretId and SecretKey in the example with your real credentials and use the current time of the system as the Timestamp.

Note: In the example below, even if you use the same programming language, the order of the parameters in the URL may be different for each execution. However, the order does not matter, as long as all the parameters are included in the URL and the signature is calculated correctly.

Note: The following code is only applicable to API 3.0. It cannot be directly used in other signature processes. Even with an older API, signature calculation errors may occur due to the differences in details. Please refer to the corresponding documentation.

Java

```
import java.nio.charset.Charset;
import java.nio.charset.StandardCharsets;
import java.security.MessageDigest;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.TimeZone;
import java.util.TreeMap;
import javax.crypto.Mac;
import javax.crypto.spec.SecretKeySpec;
import javax.xml.bind.DatatypeConverter;

public class TencentCloudAPITC3Demo {
    private final static Charset UTF8 = StandardCharsets.UTF_8;
    private final static String SECRET_ID = "AKID*****";
    private final static String SECRET_KEY = "*****";
    private final static String CT_JSON = "application/json; charset=utf-8";

    public static byte[] hmac256(byte[] key, String msg) throws Exception {
        Mac mac = Mac.getInstance("HmacSHA256");
        SecretKeySpec secretKeySpec = new SecretKeySpec(key, mac.getAlgorithm());
        mac.init(secretKeySpec);
        return mac.doFinal(msg.getBytes(UTF8));
    }

    public static String sha256Hex(String s) throws Exception {
        MessageDigest md = MessageDigest.getInstance("SHA-256");
        byte[] d = md.digest(s.getBytes(UTF8));
        return DatatypeConverter.printHexBinary(d).toLowerCase();
    }

    public static void main(String[] args) throws Exception {
        String service = "cvm";
        String host = "cvm.tencentcloudapi.com";
        String region = "ap-guangzhou";
        String action = "DescribeInstances";
        String version = "2017-03-12";
```

```

String algorithm = "TC3-HMAC-SHA256";
String timestamp = "1551113065";
//String timestamp = String.valueOf(System.currentTimeMillis() / 1000);
SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");
// Pay attention to the time zone; otherwise, errors may occur
sdf.setTimeZone(TimeZone.getTimeZone("UTC"));
String date = sdf.format(new Date(Long.valueOf(timestamp + "000")));

// ***** Step 1: Concatenate the CanonicalRequest string *****
String httpRequestMethod = "POST";
String canonicalUri = "/";
String canonicalQueryString = "";
String canonicalHeaders = "content-type:application/json; charset=utf-8\n" + "host:" + host + "\n";
String signedHeaders = "content-type;host";

String payload = "{\"Limit\": 1, \"Filters\": [{\"Values\": [\"unnamed\"], \"Name\": \"instance-name\"}] }";
String hashedRequestPayload = sha256Hex(payload);
String canonicalRequest = httpRequestMethod + "\n" + canonicalUri + "\n" + canonicalQueryString + "\n"
+ canonicalHeaders + "\n" + signedHeaders + "\n" + hashedRequestPayload;
System.out.println(canonicalRequest);

// ***** Step 2: Concatenate the string to sign *****
String credentialScope = date + "/" + service + "/" + "tc3_request";
String hashedCanonicalRequest = sha256Hex(canonicalRequest);
String stringToSign = algorithm + "\n" + timestamp + "\n" + credentialScope + "\n" + hashedCanonicalRequest;
System.out.println(stringToSign);

// ***** Step 3: Calculate the signature *****
byte[] secretDate = hmac256(("TC3" + SECRET_KEY).getBytes(UTF8), date);
byte[] secretService = hmac256(secretDate, service);
byte[] secretSigning = hmac256(secretService, "tc3_request");
String signature = DatatypeConverter.printHexBinary(hmac256(secretSigning, stringToSign)).toLowerCase();
System.out.println(signature);

// ***** Step 4: Concatenate the Authorization *****
String authorization = algorithm + " " + "Credential=" + SECRET_ID + "/" + credentialScope + ", "
+ "SignedHeaders=" + signedHeaders + ", " + "Signature=" + signature;
System.out.println(authorization);

TreeMap<String, String> headers = new TreeMap<String, String>();
headers.put("Authorization", authorization);

```

```

headers.put("Content-Type", CT_JSON);
headers.put("Host", host);
headers.put("X-TC-Action", action);
headers.put("X-TC-Timestamp", timestamp);
headers.put("X-TC-Version", version);
headers.put("X-TC-Region", region);

StringBuilder sb = new StringBuilder();
sb.append("curl -X POST https://").append(host)
.append(" -H \"Authorization: ").append(authorization).append("\")")
.append(" -H \"Content-Type: application/json; charset=utf-8\"")
.append(" -H \"Host: ").append(host).append("\")")
.append(" -H \"X-TC-Action: ").append(action).append("\")")
.append(" -H \"X-TC-Timestamp: ").append(timestamp).append("\")")
.append(" -H \"X-TC-Version: ").append(version).append("\")")
.append(" -H \"X-TC-Region: ").append(region).append("\")")
.append(" -d '").append(payload).append("'");
System.out.println(sb.toString());
}
}

```

Python

```

# -*- coding: utf-8 -*-
import hashlib, hmac, json, os, sys, time
from datetime import datetime

# Key Parameters
secret_id = "AKID*****"
secret_key = "*****"

service = "cvm"
host = "cvm.tencentcloudapi.com"
endpoint = "https://" + host
region = "ap-guangzhou"
action = "DescribeInstances"
version = "2017-03-12"
algorithm = "TC3-HMAC-SHA256"
#timestamp = int(time.time())
timestamp = 1551113065
date = datetime.utcfromtimestamp(timestamp).strftime("%Y-%m-%d")
params = {"Limit": 1, "Filters": [{"Values": ["unnamed"], "Name": "instance-name"}]}

# ***** Step 1: Concatenate the CanonicalRequest string *****
http_request_method = "POST"

```

```

canonical_uri = "/"
canonical_querystring = ""
ct = "application/json; charset=utf-8"
payload = json.dumps(params)
canonical_headers = "content-type:%s\nhost:%s\n" % (ct, host)
signed_headers = "content-type;host"
hashed_request_payload = hashlib.sha256(payload.encode("utf-8")).hexdigest()
canonical_request = (http_request_method + "\n" +
canonical_uri + "\n" +
canonical_querystring + "\n" +
canonical_headers + "\n" +
signed_headers + "\n" +
hashed_request_payload)
print(canonical_request)

# ***** Step 2: Concatenate the string to sign *****
credential_scope = date + "/" + service + "/" + "tc3_request"
hashed_canonical_request = hashlib.sha256(canonical_request.encode("utf-8")).hexdigest()
string_to_sign = (algorithm + "\n" +
str(timestamp) + "\n" +
credential_scope + "\n" +
hashed_canonical_request)
print(string_to_sign)

# ***** Step 3: Calculate the Signature *****
# Function for computing signature digest
def sign(key, msg):
return hmac.new(key, msg.encode("utf-8"), hashlib.sha256).digest()
secret_date = sign(("TC3" + secret_key).encode("utf-8"), date)
secret_service = sign(secret_date, service)
secret_signing = sign(secret_service, "tc3_request")
signature = hmac.new(secret_signing, string_to_sign.encode("utf-8"), hashlib.sha256).hexdigest()
print(signature)

# ***** Step 4: Concatenate the Authorization *****
authorization = (algorithm + " " +
"Credential=" + secret_id + "/" + credential_scope + ", " +
"SignedHeaders=" + signed_headers + ", " +
"Signature=" + signature)
print(authorization)

print('curl -X POST ' + endpoint
+ ' -H "Authorization: ' + authorization + '"
+ ' -H "Content-Type: application/json; charset=utf-8"')

```

```

+ ' -H "Host: ' + host + '" '
+ ' -H "X-TC-Action: ' + action + '" '
+ ' -H "X-TC-Timestamp: ' + str(timestamp) + '" '
+ ' -H "X-TC-Version: ' + version + '" '
+ ' -H "X-TC-Region: ' + region + '" '
+ " -d '" + payload + '"')

```

Golang

```

package main

import (
    "crypto/hmac"
    "crypto/sha256"
    "encoding/hex"
    "fmt"
    "time"
)

func sha256hex(s string) string {
    b := sha256.Sum256([]byte(s))
    return hex.EncodeToString(b[:])
}

func hmacsha256(s, key string) string {
    hashed := hmac.New(sha256.New, []byte(key))
    hashed.Write([]byte(s))
    return string(hashed.Sum(nil))
}

func main() {
    secretId := "AKID*****"
    secretKey := "*****"
    host := "cvm.tencentcloudapi.com"
    algorithm := "TC3-HMAC-SHA256"
    service := "cvm"
    version := "2017-03-12"
    action := "DescribeInstances"
    region := "ap-guangzhou"
    //var timestamp int64 = time.Now().Unix()
    var timestamp int64 = 1551113065

    // step 1: build canonical request string
    httpRequestMethod := "POST"
    canonicalURI := "/"
    canonicalQueryString := ""

```

```

canonicalHeaders := "content-type:application/json; charset=utf-8\n" + "host:" +
host + "\n"
signedHeaders := "content-type;host"
payload := `{ "Limit": 1, "Filters": [{"Values": ["unnamed"], "Name": "instance-na
me"}]}`
hashedRequestPayload := sha256hex(payload)
canonicalRequest := fmt.Sprintf("%s\n%s\n%s\n%s\n%s\n%s",
httpRequestMethod,
canonicalURI,
canonicalQueryString,
canonicalHeaders,
signedHeaders,
hashedRequestPayload)
fmt.Println(canonicalRequest)

// step 2: build string to sign
date := time.Unix(timestamp, 0).UTC().Format("2006-01-02")
credentialScope := fmt.Sprintf("%s/%s/tc3_request", date, service)
hashedCanonicalRequest := sha256hex(canonicalRequest)
string2sign := fmt.Sprintf("%s\n%d\n%s\n%s",
algorithm,
timestamp,
credentialScope,
hashedCanonicalRequest)
fmt.Println(string2sign)

// step 3: sign string
secretDate := hmacsha256(date, "TC3"+secretKey)
secretService := hmacsha256(service, secretDate)
secretSigning := hmacsha256("tc3_request", secretService)
signature := hex.EncodeToString([]byte(hmacsha256(string2sign, secretSigning)))
fmt.Println(signature)

// step 4: build authorization
authorization := fmt.Sprintf("%s Credential=%s/%s, SignedHeaders=%s, Signature=%
s",
algorithm,
secretId,
credentialScope,
signedHeaders,
signature)
fmt.Println(authorization)

curl := fmt.Sprintf(`curl -X POST https://%s\
-H "Authorization: %s"\
-H "Content-Type: application/json; charset=utf-8"\
-H "Host: %s" -H "X-TC-Action: %s"\

```

```

-H "X-TC-Timestamp: %d"\
-H "X-TC-Version: %s"\
-H "X-TC-Region: %s"\
-d '%s'`, host, authorization, host, action, timestamp, version, region, payload)
fmt.Println(curl)
}

```

PHP

```

<?php
$secretId = "AKID*****";
$secretKey = "*****";
$host = "cvm.tencentcloudapi.com";
$service = "cvm";
$version = "2017-03-12";
$action = "DescribeInstances";
$region = "ap-guangzhou";
// $timestamp = time();
$timestamp = 1551113065;
$algorithm = "TC3-HMAC-SHA256";

// step 1: build canonical request string
$httpRequestMethod = "POST";
$canonicalUri = "/";
$canonicalQueryString = "";
$canonicalHeaders = "content-type:application/json; charset=utf-8\n"."host:". $host. "\n";
$signedHeaders = "content-type;host";
$payload = '{"Limit": 1, "Filters": [{"Values": ["unnamed"], "Name": "instance-name"}]}';
$hashedRequestPayload = hash("SHA256", $payload);
$canonicalRequest = $httpRequestMethod. "\n"
.$canonicalUri. "\n"
.$canonicalQueryString. "\n"
.$canonicalHeaders. "\n"
.$signedHeaders. "\n"
.$hashedRequestPayload;
echo $canonicalRequest.PHP_EOL;

// step 2: build string to sign
$date = gmdate("Y-m-d", $timestamp);
$credentialScope = $date. "/" . $service. "/tc3_request";
$hashedCanonicalRequest = hash("SHA256", $canonicalRequest);
$stringToSign = $algorithm. "\n"
.$timestamp. "\n"
.$credentialScope. "\n"

```

```

.$hashedCanonicalRequest;
echo $stringToSign.PHP_EOL;

// step 3: sign string
$secretDate = hash_hmac("SHA256", $date, "TC3".$secretKey, true);
$secretService = hash_hmac("SHA256", $service, $secretDate, true);
$secretSigning = hash_hmac("SHA256", "tc3_request", $secretService, true);
$signature = hash_hmac("SHA256", $stringToSign, $secretSigning);
echo $signature.PHP_EOL;

// step 4: build authorization
$authorization = $algorithm
." Credential=".$secretId."/".$credentialScope
.", SignedHeaders=content-type;host, Signature=".$signature;
echo $authorization.PHP_EOL;

$curl = "curl -X POST https://" . $host
.' -H "Authorization: ' . $authorization . '"'
.' -H "Content-Type: application/json; charset=utf-8"'
.' -H "Host: ' . $host . '"'
.' -H "X-TC-Action: ' . $action . '"'
.' -H "X-TC-Timestamp: ' . $timestamp . '"'
.' -H "X-TC-Version: ' . $version . '"'
.' -H "X-TC-Region: ' . $region . '"'
." -d '" . $payload . '"';
echo $curl.PHP_EOL;

```

Ruby

```

# -*- coding: UTF-8 -*-
# require ruby>=2.3.0
require 'digest'
require 'json'
require 'time'
require 'openssl'

# Key Parameters
secret_id = 'AKID*****'
secret_key = '*****'

service = 'cvm'
host = 'cvm.tencentcloudapi.com'
endpoint = 'https://' + host
region = 'ap-guangzhou'
action = 'DescribeInstances'
version = '2017-03-12'

```

```

algorithm = 'TC3-HMAC-SHA256'
# timestamp = Time.now.to_i
timestamp = 1551113065
date = Time.at(timestamp).utc.strftime('%Y-%m-%d')

# ***** Step 1: Concatenate the CanonicalRequest string *****
http_request_method = 'POST'
canonical_uri = '/'
canonical_querystring = ''
canonical_headers = "content-type:application/json; charset=utf-8\nhost:#{host}
\n"
signed_headers = 'content-type;host'
# params = { 'Limit' => 1, 'Filters' => [{ 'Name' => 'instance-name', 'Values' =>
['unnamed'] }] }
# payload = JSON.generate(params, { 'ascii_only' => true, 'space' => ' ' })
# json will generate in random order, to get specified result in example, we hard
-code it here.
payload = '{"Limit": 1, "Filters": [{"Values": ["unnamed"], "Name": "instance-nam
e"}]}'
hashed_request_payload = Digest::SHA256.hexdigest(payload)
canonical_request = [
http_request_method,
canonical_uri,
canonical_querystring,
canonical_headers,
signed_headers,
hashed_request_payload,
].join("\n")

puts canonical_request

# ***** Step 2: Concatenate the string to sign *****
credential_scope = date + '/' + service + '/' + 'tc3_request'
hashed_request_payload = Digest::SHA256.hexdigest(canonical_request)
string_to_sign = [
algorithm,
timestamp.to_s,
credential_scope,
hashed_request_payload,
].join("\n")
puts string_to_sign

# ***** Step 3: Calculate the Signature *****
digest = OpenSSL::Digest.new('sha256')
secret_date = OpenSSL::HMAC.digest(digest, 'TC3' + secret_key, date)
secret_service = OpenSSL::HMAC.digest(digest, secret_date, service)
secret_signing = OpenSSL::HMAC.digest(digest, secret_service, 'tc3_request')

```

```

signature = OpenSSL::HMAC.hexdigest(digest, secret_signing, string_to_sign)
puts signature

# ***** Step 4: Concatenate the Authorization *****
authorization = "#{algorithm} Credential=#{secret_id}/#{credential_scope}, Signed
Headers=#{signed_headers}, Signature=#{signature}"
puts authorization

puts 'curl -X POST ' + endpoint \
+ ' -H "Authorization: ' + authorization + '" \
+ ' -H "Content-Type: application/json; charset=utf-8" \
+ ' -H "Host: ' + host + '" \
+ ' -H "X-TC-Action: ' + action + '" \
+ ' -H "X-TC-Timestamp: ' + timestamp.to_s + '" \
+ ' -H "X-TC-Version: ' + version + '" \
+ ' -H "X-TC-Region: ' + region + '" \
+ " -d '" + payload + "'"

```

DotNet

```

using System;
using System.Collections.Generic;
using System.Security.Cryptography;
using System.Text;

public class Application
{
    public static string SHA256Hex(string s)
    {
        using (SHA256 algo = SHA256.Create())
        {
            byte[] hashbytes = algo.ComputeHash(Encoding.UTF8.GetBytes(s));
            StringBuilder builder = new StringBuilder();
            for (int i = 0; i < hashbytes.Length; ++i)
            {
                builder.Append(hashbytes[i].ToString("x2"));
            }
            return builder.ToString();
        }
    }

    public static byte[] HmacSHA256(byte[] key, byte[] msg)
    {
        using (HMACSHA256 mac = new HMACSHA256(key))
        {
            return mac.ComputeHash(msg);
        }
    }
}

```

```

}

public static Dictionary<String, String> BuildHeaders(string secretid,
string secretkey, string service, string endpoint, string region,
string action, string version, DateTime date, string requestPayload)
{
    string datestr = date.ToString("yyyy-MM-dd");
    DateTime startTime = new DateTime(1970, 1, 1, 0, 0, 0, 0, DateTimeKind.Utc);
    long requestTimestamp = (long)Math.Round((date - startTime).TotalMilliseconds, MidpointRounding.AwayFromZero) / 1000;
    // ***** Step 1: Concatenate the CanonicalRequest string *****
    string algorithm = "TC3-HMAC-SHA256";
    string httpRequestMethod = "POST";
    string canonicalUri = "/";
    string canonicalQueryString = "";
    string contentType = "application/json";
    string canonicalHeaders = "content-type:" + contentType + "; charset=utf-8\n" +
"host:" + endpoint + "\n";
    string signedHeaders = "content-type;host";
    string hashedRequestPayload = SHA256Hex(requestPayload);
    string canonicalRequest = httpRequestMethod + "\n"
+ canonicalUri + "\n"
+ canonicalQueryString + "\n"
+ canonicalHeaders + "\n"
+ signedHeaders + "\n"
+ hashedRequestPayload;
    Console.WriteLine(canonicalRequest);
    Console.WriteLine("-----");

    // ***** Step 2: Concatenate the string to sign *****
    string credentialScope = datestr + "/" + service + "/" + "tc3_request";
    string hashedCanonicalRequest = SHA256Hex(canonicalRequest);
    string stringToSign = algorithm + "\n" + requestTimestamp.ToString() + "\n" + credentialScope + "\n" + hashedCanonicalRequest;
    Console.WriteLine(stringToSign);
    Console.WriteLine("-----");

    // ***** Step 3: Calculate the signature *****
    byte[] tc3SecretKey = Encoding.UTF8.GetBytes("TC3" + secretkey);
    byte[] secretDate = HmacSHA256(tc3SecretKey, Encoding.UTF8.GetBytes(datestr));
    byte[] secretService = HmacSHA256(secretDate, Encoding.UTF8.GetBytes(service));
    byte[] secretSigning = HmacSHA256(secretService, Encoding.UTF8.GetBytes("tc3_request"));
    byte[] signatureBytes = HmacSHA256(secretSigning, Encoding.UTF8.GetBytes(stringToSign));
    string signature = BitConverter.ToString(signatureBytes).Replace("-", "").ToLower();
}

```

```

Console.WriteLine(signature);
Console.WriteLine("-----");

// ***** Step 4: Concatenate the Authorization *****
string authorization = algorithm + " "
+ "Credential=" + secretid + "/" + credentialScope + ", "
+ "SignedHeaders=" + signedHeaders + ", "
+ "Signature=" + signature;
Console.WriteLine(authorization);
Console.WriteLine("-----");

Dictionary<string, string> headers = new Dictionary<string, string>();
headers.Add("Authorization", authorization);
headers.Add("Host", endpoint);
headers.Add("Content-Type", contentType + "; charset=utf-8");
headers.Add("X-TC-Timestamp", requestTimestamp.ToString());
headers.Add("X-TC-Version", version);
headers.Add("X-TC-Action", action);
headers.Add("X-TC-Region", region);
return headers;
}

public static void Main(string[] args)
{
    // SecretID and SecretKey
    string SECRET_ID = "AKID*****";
    string SECRET_KEY = "*****";

    string service = "cvm";
    string endpoint = "cvm.tencentcloudapi.com";
    string region = "ap-guangzhou";
    string action = "DescribeInstances";
    string version = "2017-03-12";

    // The timestamp `2019-02-26 00:44:25` used here is only for reference. In a project, use the following parameter:
    // DateTime date = DateTime.UtcNow;
    // Enter the correct time zone. We recommend using UTC timestamp to avoid errors.
    DateTime date = new DateTime(1970, 1, 1, 0, 0, 0, 0, DateTimeKind.Utc).AddSeconds(1551113065);
    string requestPayload = "{\"Limit\": 1, \"Filters\": [{\"Values\": [\"unnamed\"], \"Name\": \"instance-name\"}]}";

    Dictionary<string, string> headers = BuildHeaders(SECRET_ID, SECRET_KEY, service, endpoint, region, action, version, date, requestPayload);

    Console.WriteLine("POST https://cvm.tencentcloudapi.com");
    foreach (KeyValuePair<string, string> kv in headers)

```

```

{
  Console.WriteLine(kv.Key + ": " + kv.Value);
}
Console.WriteLine();
Console.WriteLine(requestPayload);
}
}

```

NodeJS

```

const crypto = require('crypto');

function sha256(message, secret = '', encoding) {
  const hmac = crypto.createHmac('sha256', secret)
  return hmac.update(message).digest(encoding)
}

function getHash(message, encoding = 'hex') {
  const hash = crypto.createHash('sha256')
  return hash.update(message).digest(encoding)
}

function getDate(timestamp) {
  const date = new Date(timestamp * 1000)
  const year = date.getUTCFullYear()
  const month = ('0' + (date.getUTCMonth() + 1)).slice(-2)
  const day = ('0' + date.getUTCDate()).slice(-2)
  return `${year}-${month}-${day}`
}

function main() {

  const SECRET_ID = "AKID*****"
  const SECRET_KEY = "*****"

  const endpoint = "cvm.tencentcloudapi.com"
  const service = "cvm"
  const region = "ap-guangzhou"
  const action = "DescribeInstances"
  const version = "2017-03-12"
  //const timestamp = getTime()
  const timestamp = 1551113065
  const date = getDate(timestamp)

  // ***** Step 1: Concatenate the CanonicalRequest string *****
  const signedHeaders = "content-type;host"

  const payload = "{\"Limit\": 1, \"Filters\": [{\"Values\": [\"unnamed\"], \"Name\": \"instance-name\"}]}"

```

```

const hashedRequestPayload = getHash(payload);
const httpRequestMethod = "POST"
const canonicalUri = "/"
const canonicalQueryString = ""
const canonicalHeaders = "content-type:application/json; charset=utf-8\n" + "host:" + endpoint + "\n"

const canonicalRequest = httpRequestMethod + "\n"
+ canonicalUri + "\n"
+ canonicalQueryString + "\n"
+ canonicalHeaders + "\n"
+ signedHeaders + "\n"
+ hashedRequestPayload
console.log(canonicalRequest)
console.log("-----")

// ***** Step 2: Concatenate the string to sign *****
const algorithm = "TC3-HMAC-SHA256"
const hashedCanonicalRequest = getHash(canonicalRequest);
const credentialScope = date + "/" + service + "/" + "tc3_request"
const stringToSign = algorithm + "\n" +
timestamp + "\n" +
credentialScope + "\n" +
hashedCanonicalRequest
console.log(stringToSign)
console.log("-----")

// ***** Step 3: Calculate the signature *****
const kDate = sha256(date, 'TC3' + SECRET_KEY)
const kService = sha256(service, kDate)
const kSigning = sha256('tc3_request', kService)
const signature = sha256(stringToSign, kSigning, 'hex')
console.log(signature)
console.log("-----")

// ***** Step 4: Concatenate the Authorization *****
const authorization = algorithm + " " +
"Credential=" + SECRET_ID + "/" + credentialScope + ", " +
"SignedHeaders=" + signedHeaders + ", " +
"Signature=" + signature
console.log(authorization)
console.log("-----")

const Call_Information = 'curl -X POST ' + "https://" + endpoint
+ ' -H "Authorization: ' + authorization + '" '
+ ' -H "Content-Type: application/json; charset=utf-8"'

```

```

+ ' -H "Host: ' + endpoint + '"
+ ' -H "X-TC-Action: ' + action + '"
+ ' -H "X-TC-Timestamp: ' + timestamp.toString() + '"
+ ' -H "X-TC-Version: ' + version + '"
+ ' -H "X-TC-Region: ' + region + '"
+ " -d '" + payload + '"
console.log(Call_Information)
}
main()

```

C++

```

#include <iostream>
#include <iomanip>
#include <sstream>
#include <string>
#include <stdio.h>
#include <time.h>
#include <openssl/sha.h>
#include <openssl/hmac.h>

using namespace std;

string get_data(int64_t &timestamp)
{
    string utcDate;
    char buff[20] = {0};
    // time_t timenow;
    struct tm sttime;
    sttime = *gmtime(&timestamp);
    strftime(buff, sizeof(buff), "%Y-%m-%d", &sttime);
    utcDate = string(buff);
    return utcDate;
}

string int2str(int64_t n)
{
    std::stringstream ss;
    ss << n;
    return ss.str();
}

string sha256Hex(const string &str)
{
    char buf[3];
    unsigned char hash[SHA256_DIGEST_LENGTH];
    SHA256_CTX sha256;
    SHA256_Init(&sha256);

```

```
SHA256_Update(&sha256, str.c_str(), str.size());
SHA256_Final(hash, &sha256);
std::string NewString = "";
for(int i = 0; i < SHA256_DIGEST_LENGTH; i++)
{
    snprintf(buf, sizeof(buf), "%02x", hash[i]);
    NewString = NewString + buf;
}
return NewString;
}
string HmacSha256(const string &key, const string &input)
{
    unsigned char hash[32];

    HMAC_CTX *h;
    #if OPENSSL_VERSION_NUMBER < 0x10100000L
    HMAC_CTX hmac;
    HMAC_CTX_init(&hmac);
    h = &hmac;
    #else
    h = HMAC_CTX_new();
    #endif

    HMAC_Init_ex(h, &key[0], key.length(), EVP_sha256(), NULL);
    HMAC_Update(h, ( unsigned char* )&input[0], input.length());
    unsigned int len = 32;
    HMAC_Final(h, hash, &len);

    #if OPENSSL_VERSION_NUMBER < 0x10100000L
    HMAC_CTX_cleanup(h);
    #else
    HMAC_CTX_free(h);
    #endif

    std::stringstream ss;
    ss << std::setfill('0');
    for (int i = 0; i < len; i++)
    {
        ss << hash[i];
    }

    return (ss.str());
}
string HexEncode(const string &input)
{
    static const char* const lut = "0123456789abcdef";
    size_t len = input.length();
```

```

string output;
output.reserve(2 * len);
for (size_t i = 0; i < len; ++i)
{
    const unsigned char c = input[i];
    output.push_back(lut[c >> 4]);
    output.push_back(lut[c & 15]);
}
return output;
}

int main()
{
    string SECRET_ID = "AKID*****";
    string SECRET_KEY = "*****";

    string service = "cvm";
    string host = "cvm.tencentcloudapi.com";
    string region = "ap-guangzhou";
    string action = "DescribeInstances";
    string version = "2017-03-12";
    int64_t timestamp = 1551113065;
    string date = get_data(timestamp);

    // ***** Step 1: Concatenate the CanonicalRequest string *****
    string httpRequestMethod = "POST";
    string canonicalUri = "/";
    string canonicalQueryString = "";
    string canonicalHeaders = "content-type:application/json; charset=utf-8\nhost:" +
    host + "\n";
    string signedHeaders = "content-type;host";
    string payload = "{\"Limit\": 1, \"Filters\": [{\"Values\": [\"unnamed\"], \"Name\": \"instance-name\"}] }";
    string hashedRequestPayload = sha256Hex(payload);
    string canonicalRequest = httpRequestMethod + "\n" + canonicalUri + "\n" + canonicalQueryString + "\n"
    + canonicalHeaders + "\n" + signedHeaders + "\n" + hashedRequestPayload;
    cout << canonicalRequest << endl;
    cout << "-----" << endl;

    // ***** Step 2: Concatenate the string to sign *****
    string algorithm = "TC3-HMAC-SHA256";
    string RequestTimestamp = int2str(timestamp);
    string credentialScope = date + "/" + service + "/" + "tc3_request";
    string hashedCanonicalRequest = sha256Hex(canonicalRequest);
    string stringToSign = algorithm + "\n" + RequestTimestamp + "\n" + credentialScop

```

```

e + "\n" + hashedCanonicalRequest;
cout << stringToSign << endl;
cout << "-----" << endl;

// ***** Step 3: Calculate the signature *****
string kKey = "TC3" + SECRET_KEY;
string kDate = HmacSha256(kKey, date);
string kService = HmacSha256(kDate, service);
string kSigning = HmacSha256(kService, "tc3_request");
string signature = HexEncode(HmacSha256(kSigning, stringToSign));
cout << signature << endl;
cout << "-----" << endl;

// ***** Step 4: Concatenate the Authorization *****
string authorization = algorithm + " " + "Credential=" + SECRET_ID + "/" + creden
tialScope + ", "
+ "SignedHeaders=" + signedHeaders + ", " + "Signature=" + signature;
cout << authorization << endl;
cout << "-----" << endl;

string headers = "curl -X POST https://" + host + "\n"
+ " -H \"Authorization: \" + authorization + "\n"
+ " -H \"Content-Type: application/json; charset=utf-8\" + "\n"
+ " -H \"Host: \" + host + "\n"
+ " -H \"X-TC-Action: \" + action + "\n"
+ " -H \"X-TC-Timestamp: \" + RequestTimestamp + "\n"
+ " -H \"X-TC-Version: \" + version + "\n"
+ " -H \"X-TC-Region: \" + region + "\n"
+ " -d '" + payload;
cout << headers << endl;
return 0;
};

```

Signature Failure

The following situational error codes for signature failure may occur. Please resolve the errors accordingly.

Error Code	Description
AuthFailure.SignatureExpire	Signature expired. Timestamp and server time cannot differ by more than five minutes.
AuthFailure.SecretIdNotFound	The key does not exist. Please go to the console to check whether it is disabled or you copied fewer or more characters.

AuthFailure.SignatureFailure	Signature error. It is possible that the signature was calculated incorrectly, the signature does not match the content actually sent, or the SecretKey is incorrect.
AuthFailure.TokenFailure	Temporary certificate token error.
AuthFailure.InvalidSecretId	Invalid key (not a TencentCloud API key type).

Signature

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Tencent Cloud API authenticates each access request, i.e. each request needs to include authentication information (Signature) in the common parameters to verify the identity of the requester.

The Signature is generated by the security credentials which include SecretId and SecretKey. If you don't have the security credentials yet, go to the [TencentCloud API Key](#) page to apply for them; otherwise, you cannot invoke the TencentCloud API.

1. Applying for Security Credentials

Before using the TencentCloud API for the first time, go to the [TencentCloud API Key](#) page to apply for security credentials.

Security credentials consist of SecretId and SecretKey:

- SecretId is used to identify the API requester.
- SecretKey is used to encrypt the signature string and verify it on the server.
- **You must keep your security credentials private and avoid disclosure.**

You can apply for the security credentials through the following steps:

1. Log in to the [Tencent Cloud Console](#).
2. Go to the [TencentCloud API Key](#) page.
3. On the [API Key Management](#) page, click **Create Key** to create a SecretId/SecretKey pair.

Note: Each account can have up to two pairs of SecretId/SecretKey.

2. Generating a Signature

With the SecretId and SecretKey, a signature can be generated. The following describes how to generate a signature:

Assume that the SecretId and SecretKey are:

- SecretId: AKID*****
- SecretKey: *****

Note: This is just an example. For actual operations, please use your own SecretId and SecretKey.

Take the Cloud Virtual Machine's request to view the instance list (DescribeInstances) as an example. When you invoke this API, the request parameters may be as follows:

Parameter name	Description	Parameter value
Action	Method name	DescribeInstances
SecretId	Key ID	AKID*****
Timestamp	Current timestamp	1465185768
Nonce	Random positive integer	11886
Region	Region where the instance is located	ap-guangzhou
InstanceIds.0	ID of the instance to query	ins-09dx96dg
Offset	Offset	0
Limit	Allowed maximum output	20
Version	API version number	2017-03-12

2.1. Sorting Parameters

First, sort all the request parameters in an ascending lexicographical order (ASCII code) by their names.

Notes: (1) Parameters are sorted by their names instead of their values; (2) The parameters are sorted based on ASCII code, not in an alphabetical order or by values. For example, InstanceIds.2 should be arranged after InstanceIds.12. You can complete the sorting process using a sorting function in a programming language, such as the ksort function in PHP. The parameters in the example are sorted as follows:

```
{
  'Action' : 'DescribeInstances',
  'InstanceIds.0' : 'ins-09dx96dg',
  'Limit' : 20,
  'Nonce' : 11886,
  'Offset' : 0,
  'Region' : 'ap-guangzhou',
  'SecretId' : 'AKID*****',
  'Timestamp' : 1465185768,
  'Version' : '2017-03-12',
}
```

When developing in another programming language, you can sort these sample parameters and it will work as long as you obtain the same results.

2.2. Concatenating a Request String

This step generates a request string.

Format the request parameters sorted in the previous step into the form of "parameter name"="parameter value". For example, for the Action parameter, its parameter name is "Action" and its parameter value is "DescribeInstances", so it will become Action=DescribeInstances after formatted.

Note: The "parameter value" is the original value but not the value after URL encoding.

Then, concatenate the formatted parameters with "&". The resulting request string is as follows:

```
Action=DescribeInstances&InstanceIds.0=ins-09dx96dg&Limit=20&Nonce=11886&Offset=0
&Region=ap-guangzhou&SecretId=AKID*****&Timestamp=1465
185768&Version=2017-03-12
```

2.3. Concatenating the Signature Original String

This step generates a signature original string.

The signature original string consists of the following parameters:

1. HTTP method: POST and GET modes are supported, and GET is used here for the request. Please note that the method name should be in all capital letters.
2. Request server: the domain name of the request to view the list of instances (DescribeInstances) is cvm.tencentcloudapi.com. The actual request domain name varies by the module to which the API belongs. For more information, see the instructions of the specific API.
3. Request path: The request path in the current version of TencentCloud API is fixed to /.
4. Request string: the request string generated in the previous step.

The concatenation rule of the signature original string is: Request method + request host + request path + ? + request string

The concatenation result of the example is:

```
GETcvm.tencentcloudapi.com/?Action=DescribeInstances&InstanceIds.0=ins-09dx96dg&L
imit=20&Nonce=11886&Offset=0&Region=ap-guangzhou&SecretId=AKID*****
*****&Timestamp=1465185768&Version=2017-03-12
```

2.4. Generating a Signature String

This step generates a signature string.

First, use the HMAC-SHA1 algorithm to sign the **signature original string** obtained in the previous step, and

then encode the generated signature using Base64 to obtain the final signature.

The specific code is as follows with the PHP language being used as an example:

```
$secretKey = '*****';
$srcStr = 'GETcvm.tencentcloudapi.com/?Action=DescribeInstances&InstanceIds.0=ins-09dx96dg&Limit=20&Nonce=11886&Offset=0&Region=ap-guangzhou&SecretId=AKID*****&Timestamp=1465185768&Version=2017-03-12';
$signStr = base64_encode(hash_hmac('sha1', $srcStr, $secretKey, true));
echo $signStr;
```

The final signature is:

```
7RAM2xfNMO9EiVTNmPg06MRnCvQ=
```

When developing in another programming language, you can sign and verify the original in the example above and it works as long as you get the same results.

3. Encoding a Signature String

The generated signature string cannot be directly used as a request parameter and must be URL encoded.

For example, if the signature string generated in the previous step is 7RAM2xfNMO9EiVTNmPg06MRnCvQ=, the final signature string request parameter (Signature) is 7RAM2xfNMO9EiVTNmPg06MRnCvQ%3D, which will be used to generate the final request URL.

Note: If your request method is GET, or the request method is POST and the Content-Type is application/x-www-form-urlencoded, then all the request parameter values need to be URL encoded (except the parameter key and the symbol of =) when sending the request. Non-ASCII characters need to be encoded with UTF-8 before URL encoding.

Note: The network libraries of some programming languages automatically URL encode all parameters, in which case there is no need to URL encode the signature string; otherwise, two rounds of URL encoding will cause the signature to fail.

Note: Other parameter values also need to be encoded using [RFC 3986](#). Use %XY in percent-encoding for special characters such as Chinese characters, where "X" and "Y" are hexadecimal characters (0–9 and uppercase A–F), and using lowercase will cause an error.

4. Signature Failure

The following situational error codes for signature failure may occur. Please resolve the errors accordingly.

Error code	Error description
AuthFailure.SignatureExpire	The signature is expired
AuthFailure.SecretIdNotFound	The key does not exist
AuthFailure.SignatureFailure	Signature error
AuthFailure.TokenFailure	Token error
AuthFailure.InvalidSecretId	Invalid key (not a TencentCloud API key type)

5. Signature Demo

When calling API 3.0, you are recommended to use the corresponding Tencent Cloud SDK 3.0 which encapsulates the signature process, enabling you to focus on only the specific APIs provided by the product when developing. See [SDK Center](#) for more information. Currently, the following programming languages are supported:

- [Python](#)
- [Java](#)
- [PHP](#)
- [Go](#)
- [NodeJS](#)
- [.NET](#)

To further explain the signing process, we will use a programming language to implement the process described above. The request domain name, API and parameter values in the sample are used here. This goal of this example is only to provide additional clarification for the signature process, please see the SDK for actual usage.

The final output URL might be: `https://cvm.tencentcloudapi.com/?Action=DescribeInstances&InstanceIds.0=ins-09dx96dg&Limit=20&Nonce=11886&Offset=0&Region=ap-guangzhou&SecretId=AKID*****&Signature=7RAM2xfNM09EiVTNmPg06MRnCvQ%3D&Timestamp=1465185768&Version=2017-03-12` .

Note: The key in the example is fictitious, and the timestamp is not the current time of the system, so if this URL is opened in the browser or called using commands such as curl, an authentication error will be

returned: Signature expired. In order to get a URL that can work properly, you need to replace the SecretId and SecretKey in the example with your real credentials and use the current time of the system as the Timestamp.

Note: In the example below, even if you use the same programming language, the order of the parameters in the URL may be different for each execution. However, the order does not matter, as long as all the parameters are included in the URL and the signature is calculated correctly.

Note: The following code is only applicable to API 3.0. It cannot be directly used in other signature processes. Even with an older API, signature calculation errors may occur due to the differences in details. Please refer to the corresponding documentation.

Java

```
import java.io.UnsupportedEncodingException;
import java.net.URLEncoder;
import java.util.Random;
import java.util.TreeMap;
import javax.crypto.Mac;
import javax.crypto.spec.SecretKeySpec;
import javax.xml.bind.DatatypeConverter;

public class TencentCloudAPIDemo {
    private final static String CHARSET = "UTF-8";

    public static String sign(String s, String key, String method) throws Exception {
        Mac mac = Mac.getInstance(method);
        SecretKeySpec secretKeySpec = new SecretKeySpec(key.getBytes(CHARSET), mac.getAlgorithm());
        mac.init(secretKeySpec);
        byte[] hash = mac.doFinal(s.getBytes(CHARSET));
        return DatatypeConverter.printBase64Binary(hash);
    }

    public static String getStringToSign(TreeMap<String, Object> params) {
        StringBuilder s2s = new StringBuilder("GETcvm.tencentcloudapi.com/?");
        // When signing, the parameters need to be sorted in lexicographical order. TreeMap
        // is used here to guarantee the correct order.
        for (String k : params.keySet()) {
            s2s.append(k).append("=").append(params.get(k).toString()).append("&");
        }
        return s2s.toString().substring(0, s2s.length() - 1);
    }

    public static String getUrl(TreeMap<String, Object> params) throws UnsupportedEncodingException {
```

```

StringBuilder url = new StringBuilder("https://cvm.tencentcloudapi.com/?");
// There is no requirement for the order of the parameters in the actual request
URL.
for (String k : params.keySet()) {
    // The request string needs to be URL encoded. As the Key is all in English letters,
    // only the value is URL encoded here.
    url.append(k).append("=").append(URLEncoder.encode(params.get(k).toString(), CHARSET))
        .append("&");
}
return url.toString().substring(0, url.length() - 1);
}

public static void main(String[] args) throws Exception {
    TreeMap<String, Object> params = new TreeMap<String, Object>(); // TreeMap enables
    // automatic sorting
    // A random number should be used when actually calling, for example: params.put
    // ("Nonce", new Random().nextInt(Integer.MAX_VALUE));
    params.put("Nonce", 11886); // Common parameter
    // The current time of the system should be used when actually calling, for example:
    // params.put("Timestamp", System.currentTimeMillis() / 1000);
    params.put("Timestamp", 1465185768); // Common parameter
    params.put("SecretId", "AKID*****"); // Common parameter
    params.put("Action", "DescribeInstances"); // Common parameter
    params.put("Version", "2017-03-12"); // Common parameter
    params.put("Region", "ap-guangzhou"); // Common parameter
    params.put("Limit", 20); // Business parameter
    params.put("Offset", 0); // Business parameter
    params.put("InstanceIds.0", "ins-09dx96dg"); // Business parameter
    params.put("Signature", sign(getStringToSign(params), "*****"
    + "*****", "HmacSHA1")); // Common parameter
    System.out.println(getUrl(params));
}
}

```

Python

Note: If running in a Python 2 environment, the following requests dependency package must be installed first: `pip install requests`.

```

# -*- coding: utf8 -*-
import base64
import hashlib
import hmac
import time

```

```

import requests

secret_id = "AKID*****"
secret_key = "*****"

def get_string_to_sign(method, endpoint, params):
    s = method + endpoint + "/"
    query_str = "&".join("%s=%s" % (k, params[k]) for k in sorted(params))
    return s + query_str

def sign_str(key, s, method):
    hmac_str = hmac.new(key.encode("utf8"), s.encode("utf8"), method).digest()
    return base64.b64encode(hmac_str)

if __name__ == '__main__':
    endpoint = "cvm.tencentcloudapi.com"
    data = {
        'Action': 'DescribeInstances',
        'InstanceIds.0': 'ins-09dx96dg',
        'Limit': 20,
        'Nonce': 11886,
        'Offset': 0,
        'Region': 'ap-guangzhou',
        'SecretId': secret_id,
        'Timestamp': 1465185768, # int(time.time())
        'Version': '2017-03-12'
    }
    s = get_string_to_sign("GET", endpoint, data)
    data["Signature"] = sign_str(secret_key, s, hashlib.sha1)
    print(data["Signature"])
    # An actual invocation would occur here, which may incur fees after success
    # resp = requests.get("https://" + endpoint, params=data)
    # print(resp.url)

```

Golang

```

package main

import (
    "bytes"
    "crypto/hmac"
    "crypto/sha1"
    "encoding/base64"
    "fmt"
    "sort"
)

```

```

func main() {
secretId := "AKID*****"
secretKey := "*****"
params := map[string]string{
    "Nonce": "11886",
    "Timestamp": "1465185768",
    "Region": "ap-guangzhou",
    "SecretId": secretId,
    "Version": "2017-03-12",
    "Action": "DescribeInstances",
    "InstanceIds.0": "ins-09dx96dg",
    "Limit": "20",
    "Offset": "0",
}

var buf bytes.Buffer
buf.WriteString("GET")
buf.WriteString("cvm.tencentcloudapi.com")
buf.WriteString("/")
buf.WriteString("?")

// sort keys by ascii asc order
keys := make([]string, 0, len(params))
for k, _ := range params {
    keys = append(keys, k)
}
sort.Strings(keys)

for i := range keys {
    k := keys[i]
    buf.WriteString(k)
    buf.WriteString("=")
    buf.WriteString(params[k])
    buf.WriteString("&")
}
buf.Truncate(buf.Len() - 1)

hashed := hmac.New(sha1.New, []byte(secretKey))
hashed.Write(buf.Bytes())

fmt.Println(base64.StdEncoding.EncodeToString(hashed.Sum(nil)))
}

```

PHP

```

<?php
$secretId = "AKID*****";
$secretKey = "*****";
$params["Nonce"] = 11886;//rand();
$params["Timestamp"] = 1465185768;//time();
$params["Region"] = "ap-guangzhou";
$params["SecretId"] = $secretId;
$params["Version"] = "2017-03-12";
$params["Action"] = "DescribeInstances";
$params["InstanceIds.0"] = "ins-09dx96dg";
$params["Limit"] = 20;
$params["Offset"] = 0;

ksort($params);

$signStr = "GETcvm.tencentcloudapi.com/?";
foreach ( $params as $key => $value ) {
    $signStr = $signStr . $key . "=" . $value . "&";
}
$signStr = substr($signStr, 0, -1);

$signature = base64_encode(hash_hmac("sha1", $signStr, $secretKey, true));
echo $signature.PHP_EOL;
// need to install and enable curl extension in php.ini
// $params["Signature"] = $signature;
// $url = "https://cvm.tencentcloudapi.com/?".http_build_query($params);
// echo $url.PHP_EOL;
// $ch = curl_init();
// curl_setopt($ch, CURLOPT_URL, $url);
// $output = curl_exec($ch);
// curl_close($ch);
// echo json_decode($output);

```

Ruby

```

# -*- coding: UTF-8 -*-
# require ruby>=2.3.0
require 'time'
require 'openssl'
require 'base64'

secret_id = "AKID*****"
secret_key = "*****"

method = 'GET'

```

```

endpoint = 'cvm.tencentcloudapi.com'
data = {
  'Action' => 'DescribeInstances',
  'InstanceIds.0' => 'ins-09dx96dg',
  'Limit' => 20,
  'Nonce' => 11886,
  'Offset' => 0,
  'Region' => 'ap-guangzhou',
  'SecretId' => secret_id,
  'Timestamp' => 1465185768, # Time.now.to_i
  'Version' => '2017-03-12',
}
sign = method + endpoint + '/?'
params = []
data.sort.each do |item|
  params << "#{item[0]}=#{item[1]}"
end
sign += params.join('&')
digest = OpenSSL::Digest.new('sha1')
data['Signature'] = Base64.encode64(OpenSSL::HMAC.digest(digest, secret_key, sign))
puts data['Signature']

# require 'net/http'
# uri = URI('https://' + endpoint)
# uri.query = URI.encode_www_form(data)
# p uri
# res = Net::HTTP.get_response(uri)
# puts res.body

```

DotNet

```

using System;
using System.Collections.Generic;
using System.Net;
using System.Security.Cryptography;
using System.Text;

public class Application {
  public static string Sign(string signKey, string secret)
  {
    string signRet = string.Empty;
    using (HMACSHA1 mac = new HMACSHA1(Encoding.UTF8.GetBytes(signKey)))
    {
      byte[] hash = mac.ComputeHash(Encoding.UTF8.GetBytes(secret));
      signRet = Convert.ToBase64String(hash);
    }
  }
}

```

```

}
return signRet;
}

public static string MakeSignPlainText(SortedDictionary<string, string> requestParams, string requestMethod, string requestHost, string requestPath)
{
    string retStr = "";
    retStr += requestMethod;
    retStr += requestHost;
    retStr += requestPath;
    retStr += "?";
    string v = "";
    foreach (string key in requestParams.Keys)
    {
        v += string.Format("{0}={1}&", key, requestParams[key]);
    }
    retStr += v.TrimEnd('&');
    return retStr;
}

public static void Main(string[] args)
{
    string SECRET_ID = "AKID*****";
    string SECRET_KEY = "*****";

    string endpoint = "cvm.tencentcloudapi.com";
    string region = "ap-guangzhou";
    string action = "DescribeInstances";
    string version = "2017-03-12";
    double RequestTimestamp = 1465185768;
    // long timestamp = ToTimestamp() / 1000;
    // string requestTimestamp = timestamp.ToString();
    Dictionary<string, string> param = new Dictionary<string, string>();
    param.Add("Limit", "20");
    param.Add("Offset", "0");
    param.Add("InstanceIds.0", "ins-09dx96dg");
    param.Add("Action", action);
    param.Add("Nonce", "11886");
    // param.Add("Nonce", Math.Abs(new Random().Next()).ToString());

    param.Add("Timestamp", RequestTimestamp.ToString());
    param.Add("Version", version);

    param.Add("SecretId", SECRET_ID);
    param.Add("Region", region);
    SortedDictionary<string, string> headers = new SortedDictionary<string, string>(param, StringComparer.Ordinal);

```

```

string sigInParam = MakeSignPlainText(headers, "GET", endpoint, "/");
Console.WriteLine(sigInParam);
string sigOutParam = Sign(SECRET_KEY, sigInParam);

Console.WriteLine("GET https://cvm.tencentcloudapi.com");
foreach (KeyValuePair<string, string> kv in headers)
{
    Console.WriteLine(kv.Key + ": " + kv.Value);
}
Console.WriteLine("Signature" + ": " + WebUtility.UrlEncode(sigOutParam));
Console.WriteLine();

string result = "https://cvm.tencentcloudapi.com/?";
foreach (KeyValuePair<string, string> kv in headers)
{
    result += WebUtility.UrlEncode(kv.Key) + "=" + WebUtility.UrlEncode(kv.Value) +
    "&";
}
result += WebUtility.UrlEncode("Signature") + "=" + WebUtility.UrlEncode(sigOutPa
ram);
Console.WriteLine("GET " + result);
}
}

```

NodeJS

```

const crypto = require('crypto');

function get_req_url(params, endpoint){
    params['Signature'] = escape(params['Signature']);
    const url_strParam = sort_params(params)
    return "https://" + endpoint + "/" + "?" + url_strParam.slice(1);
}

function formatSignString(reqMethod, endpoint, path, strParam){
    let strSign = reqMethod + endpoint + path + "?" + strParam.slice(1);
    return strSign;
}

function sha1(secretKey, strsign){
    let signMethodMap = {'HmacSHA1': "sha1"};
    let hmac = crypto.createHmac(signMethodMap['HmacSHA1'], secretKey || "");
    return hmac.update(Buffer.from(strsign, 'utf8')).digest('base64')
}

function sort_params(params) {

```

```
let strParam = "";
let keys = Object.keys(params);
keys.sort();
for (let k in keys) {
  //k = k.replace(/_/g, '.');
  strParam += ("&" + keys[k] + "=" + params[keys[k]]);
}
return strParam
}

function main() {
const SECRET_ID = "AKID*****"
const SECRET_KEY = "*****"

const endpoint = "cvm.tencentcloudapi.com"
const Region = "ap-guangzhou"
const Version = "2017-03-12"
const Action = "DescribeInstances"
const Timestamp = 1465185768
// const Timestamp = Math.round(Date.now() / 1000)
const Nonce = 11886
//const nonce = Math.round(Math.random() * 65535)

let params = {};
params['Action'] = Action;
params['InstanceIds.0'] = 'ins-09dx96dg';
params['Limit'] = 20;
params['Offset'] = 0;
params['Nonce'] = Nonce;
params['Region'] = Region;
params['SecretId'] = SECRET_ID;
params['Timestamp'] = Timestamp;
params['Version'] = Version;

strParam = sort_params(params)

const reqMethod = "GET";
const path = "/";
strSign = formatSignString(reqMethod, endpoint, path, strParam)
console.log(strSign)
console.log("-----")

params['Signature'] = sha1(SECRET_KEY, strSign)
console.log(params['Signature'])
console.log("-----")

const req_url = get_req_url(params, endpoint)
```

```
console.log(params['Signature'])
console.log("-----")
console.log(req_url)
}
main()
```

Responses

Last updated: 2026-02-13 16:53:09

Response for Successful Requests

For example, when calling CAM API (version: 2017-03-12) to view the status of instances (DescribeInstancesStatus), if the request has succeeded, you may see the response as shown below:

```
{
  "Response": {
    "TotalCount": 0,
    "InstanceStatusSet": [],
    "RequestId": "b5b41468-520d-4192-b42f-595cc34b6c1c"
  }
}
```

- The API will return `Response`, which contains `RequestId`, as long as it processes the request. It does not matter if the request is successful or not.
- RequestId is the unique ID of an API request. Contact us with this ID when an exception occurs.
- Except for the fixed fields, all fields are action-specified. For the definitions of action-specified fields, see the corresponding API documentation. In this example, `TotalCount` and `InstanceStatusSet` are the fields specified by the API `DescribeInstancesStatus`. `0` `TotalCount` means that the requester owns 0 CVM instance so the `InstanceStatusSet` is empty.

Response for Failed Requests

If the request has failed, you may see the response as shown below:

```
{
  "Response": {
    "Error": {
      "Code": "AuthFailure.SignatureFailure",
      "Message": "The provided credentials could not be validated. Please ensure your signature is correct."
    },
    "RequestId": "ed93f3cb-f35e-473f-b9f3-0d451b8b79c6"
  }
}
```

- The presence of the `Error` field indicates that the request has failed. A response for a failed request will include `Error`, `Code` and `Message` fields.
- `Code` is the code of the error that helps you identify the cause and solution. There are two types of error codes so you may find the code in either common error codes or API-specified error codes.
- `Message` explains the cause of the error. Note that the returned messages are subject to service updates. The information the messages provide may not be up-to-date and should not be the only source of reference.
- `RequestId` is the unique ID of an API request. Contact us with this ID when an exception occurs.

Common Error Codes

If there is an `Error` field in the response, it means that the API call failed. The `Code` field in `Error` indicates the error code. The following table lists the common error codes that all actions can return.

Error Code	Description
<code>AuthFailure.InvalidSecretId</code>	Invalid key (not a TencentCloud API key type).
<code>AuthFailure.MFAFailure</code>	MFA failed.
<code>AuthFailure.SecretIdNotFound</code>	The key does not exist.
<code>AuthFailure.SignatureExpire</code>	Signature expired.
<code>AuthFailure.SignatureFailure</code>	Signature error.
<code>AuthFailure.TokenFailure</code>	Token error.
<code>AuthFailure.UnauthorizedOperation</code>	The request does not have CAM authorization.
<code>DryRunOperation</code>	DryRun Operation. It means that the request would have succeeded, but the <code>DryRun</code> parameter was used.
<code>FailedOperation</code>	Operation failed.
<code>InternalError</code>	Internal error.
<code>InvalidAction</code>	The API does not exist.
<code>InvalidParameter</code>	Incorrect parameter.
<code>InvalidParameterValue</code>	Invalid parameter value.
<code>LimitExceeded</code>	Quota limit exceeded.

MissingParameter	A parameter is missing.
NoSuchVersion	The API version does not exist.
RequestLimitExceeded	The number of requests exceeds the frequency limit.
ResourceInUse	Resource is in use.
ResourceInsufficient	Insufficient resource.
ResourceNotFound	The resource does not exist.
ResourceUnavailable	Resource is unavailable.
UnauthorizedOperation	Unauthorized operation.
UnknownParameter	Unknown parameter.
UnsupportedOperation	Unsupported operation.
UnsupportedProtocol	HTTPS request method error. Only GET and POST requests are supported.
UnsupportedRegion	API does not support the requested region.

Hunyuan Text APIs

ChatTranslations

Last updated: 2026-01-27 12:37:24

1. API Description

Domain name for API request: hunyuan.ai.intl.tencentcloudapi.com.

Tencent Hunyuan is a large language model (LLM) developed by Tencent R&D. It possesses powerful Chinese creation capacity, logical reasoning in complex context, and reliable task execution power. This API supports streaming or non-streaming calls. When using streaming calls, it follows the SSE protocol.

1. This API does not currently support returning Image Content.
2. By default, this API has account restrictions with a number of concurrencies of 5.
3. Please use the SDK to call this API. examples are provided in the Git repository [examples/hunyuan/v20230901/](#) directory for each development language. The SDK link is provided in the **"Developer Resources – SDK"** part under the document.
4. We recommend you use API Explorer for quick online debugging interface and download example code in languages, [click to open](#).

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API:

			ChatTranslations.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	No	String	Common Params . This parameter is not required.
Model	Yes	String	Model name. optional values include hunyuan-translation. Please read the introduction in the product overview for model descriptions. Note: Different models have different pricing. according to the purchase guide, call as needed.
Stream	No	Boolean	Streaming call switch. Note:. 1. it defaults to non-streaming (false) when no value is passed. 2. for streaming calls, the results are incrementally returned via the SSE protocol (the return value is taken from Choices[n].Delta, and incremental data must be concatenated to obtain the complete result). 3. for non-streaming calls: The calling method is the same as an ordinary HTTP request. The API response is time-consuming. if needed, set it to true for reduced latency. Only return the final result once (return value takes the value from Choices[n].Message). Note:. When making an SDK call, streaming and non-streaming calls require different ways to obtain the return value. refer to the comments or sample code in the SDK (in the examples/hunyuan/v20230901/ directory of each language SDK code repository).
Text	No	String	Text to be translated.
Source	No	String	Source language. Supported language list:. Simplified chinese: zh, traditional chinese: zh-TR, cantonese: yue, english: en, french: fr, portuguese: pt, spanish: es, japanese: ja, turkish: TR, russian: ru, arabic: ar, korean: ko, thai: th, italian: it, german: de, vietnamese: vi, malay: ms, indonesian: id.

			The following languages are supported only by the hunyuan–translation model:. Filipino: fil, hindi: hi, polish: pl, czech: cs, dutch: nl, khmer: km, burmese: my, persian: fa, gujarati: gu, urdu: ur, telugu: te, marathi: mr, hebrew: he, bengali: bn, tamil: ta, ukrainian: uk, tibetan: bo, kazakh: kk, mongolian: mn, uyghur: ug.
Target	No	String	Target language. Supported language list:. Simplified chinese: zh, traditional chinese: zh–TR, cantonese: yue, english: en, french: fr, portuguese: pt, spanish: es, japanese: ja, turkish: TR, russian: ru, arabic: ar, korean: ko, thai: th, italian: it, german: de, vietnamese: vi, malay: ms, indonesian: id. The following languages are supported only by the hunyuan–translation model:. Filipino: fil, hindi: hi, polish: pl, czech: cs, dutch: nl, khmer: km, burmese: my, persian: fa, gujarati: gu, urdu: ur, telugu: te, marathi: mr, hebrew: he, bengali: bn, tamil: ta, ukrainian: uk, tibetan: bo, kazakh: kk, mongolian: mn, uyghur: ug.
Field	No	String	Domain of the text to be translated, such as game plot.
References.N	No	Array of Reference	Reference example, up to 10.

3. Output Parameters

Parameter Name	Type	Description
Id	String	Request'S RequestId this time.
Note	String	Disclaimer.
Created	Integer	Unix timestamp, in seconds.
Usage	Usage	Token statistical information. Billing by Token quantity.
Choices	Array of TranslationChoice	Reply content.

ErrorMsg	ErrorMsg	Error message. If the service encounters an exception during streaming return, return this error.
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem. As a streaming response API, when the request is successfully completed, the RequestId will be placed in the Header "X-TC-RequestId" of the HTTP response.

4. Example

Example1 Non-Streaming Call to Translation API

Non-streaming call to Translation API

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: ChatTranslations
<Common request parameters>

{
  "Model": "hunyuan-translation",
  "Stream": false,
  "Text": "Playing video games leads to lasting improvements in cognitive abilities.",
  "Source": "en",
  "Target": "zh",
  "Field": "Game plot"
  "References": [
    {
      "Type": "sentence",
      "Text": "Computer games are a perfect recipe for strengthening our cognitive skills",
      "Translation": "Computer games are the perfect recipe to enhance our cognition"
    }
  ]
}
```

Output Example

```
{
  "Response": {
    "RequestId": "e638f0a0-a2a5-4c43-a1c6-798a7cbfbb18",
    "Id": "e638f0a0-a2a5-4c43-a1c6-798a7cbfbb18",
    "Created": 1733133524,
    "Choices": [
      {
        "Index": 0,
        "Message": {
          "Role": "assistant",
          "Content": "Playing video games can cause persistent enhancement in cognition."
        },
        "FinishReason": "stop"
      }
    ],
    "Usage": {
      "PromptTokens": 96,
      "CompletionTokens": 10,
      "TotalTokens": 106
    },
    "Note": "The above content is AI generation and does not mean the developer stand point. Do not delete or modify this tag."
  }
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

- [Tencent Cloud SDK 3.0 for Python](#)
- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

The following only lists the error codes related to the API business logic. For other error codes, see [Common Error Codes](#).

Error Code	Description
FailedOperation.EngineRequestTimeout	Request timeout at the engine layer. try again later.
FailedOperation.EngineServerError	Internal error at the engine layer. try again later.
FailedOperation.EngineServerLimitExceeded	Request exceeds the limit at the engine layer. try again later.
FailedOperation.FreeResourcePackExhausted	The remaining free resource package quota is exhausted. purchase a resource pack or enable post-payment.
FailedOperation.ResourcePackExhausted	The remaining resource package quota is exhausted. purchase a resource pack or enable post-payment.
FailedOperation.ServiceNotActivated	The service is not activated. go to the console to apply for a trial.
FailedOperation.ServiceStop	User proactively stopped services.
FailedOperation.ServiceStopArrears	Service suspended due to overdue payments.
InternalError	Internal error.
InvalidParameter	Parameter error.
InvalidParameterValue	Parameter value error.
InvalidParameterValue.Model	Model does not exist.
LimitExceeded	The quota limit is exceeded.

Hunyuan 3D APIs

SubmitHunyuanTo3DProJob

Last updated: 2026-01-27 14:54:56

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

This API is used to intelligently generate 3D content based on the HunYuan Large Model and input text descriptions/images.

This API is used to provide 3 concurrent tasks by default. Up to 3 submitted tasks can be processed simultaneously. A new task can be processed only after the previous one is completed.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: SubmitHunyuanTo3DProJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
Model	No	String	Tencent HY 3D Global model version Defaults to 3.0, with optional choices: 3.0, 3.1 When selecting version 3.1, the [LowPoly] and

			[Sketch] parameter is unavailable Example value:3.0
Prompt	No	String	Generates 3D content, describes 3D content. Supports up to 1024 utf-8 characters. Text-To-3D. Specifies either ImageBase64/ImageUrl or Prompt is required. Prompt and ImageBase64/ImageUrl cannot coexist.
ImageBase64	No	String	Enter the Base64 code of the image. Size: specifies the unilateral resolution requirement, not less than 128 and not greater than 5000. size should not exceed 8m (after encoding, it increases by about 30%, recommend actual input image size no more than 6m). Input image suggestion: 1.Simple background (solid-color background) 2.No text or blended colors 3.Single object 4.The object occupies over 50% of the frame Valid values: jpg, png, jpeg, webp. Specifies either ImageBase64/ImageUrl or Prompt is required. Prompt and ImageBase64/ImageUrl cannot coexist.
ImageUrl	No	String	Input image Url. Size: specifies the unilateral resolution requirement, not less than 128 and not greater than 5000. size should not exceed 8m (after encoding, it increases by about 30%, recommend actual input image size no more than 6m). Input image suggestion: 1.Simple background (solid-color background) 2.No text or blended colors 3.Single object 4.The object occupies over 50% of the frame Valid values: jpg, png, jpeg, webp. Specifies either ImageBase64/ImageUrl or Prompt is required. Prompt and ImageBase64/ImageUrl cannot coexist.
MultiViewImages.N	No	Array of ViewImage	Multi-Perspective model image. reference value for viewing angle: left: Left view; right: Right view; back: Rear view;

			top: Top view (only supported in Model 3.1); bottom: Bottom view (only supported in Model 3.1); left_front: Left front 45 degree view (only supported in Model 3.1); right_front: Right front 45 degree view (only supported in Model 3.1); Each perspective is limited to one image. Image size limit. the value must not exceed 8 mb after encoding. Image resolution limitation: the unilateral resolution should be less than 5000 and greater than 128. Supported image format: JPG or PNG
EnablePBR	No	Boolean	Specifies whether PBR material generation is enabled. default false
FaceCount	No	Integer	Specifies the face count for 3D model generation. default value is 500000. Specifies the supported face count generation range. value range: 40000–1500000
GenerateType	No	String	Generation task type. default: Normal. valid values: Normal: generates a geometric model with textures LowPoly: specifies the model generated after intelligent polygon reduction. Geometry: specifies whether to generate a Geometry model without textures (white model). when this task is selected, the EnablePBR parameter does not take effect Specifies the Sketch for the generative model, allowing input of a Sketch or line drawing. in this mode, both prompt and ImageUrl/ImageBase64 can be entered together
PolygonType	No	String	This parameter only takes effect when LowPoly mode is selected from GenerateType Polygon type, indicates the number of sides in the model's surface grid, defaults to triangle. valid values: triangle. specifies the triangular face quadrilateral: mix quadrangle and triangle faces to generate

3. Output Parameters

Parameter Name	Type	Description
JobId	String	Task ID (valid period: 24 hours).
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Submitting a 3D Task

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: SubmitHunyuanTo3DProJob
<Common request parameters>

{
  "Prompt": "A cat"
}
```

Output Example

```
{
  "Response": {
    "JobId": "137536775519696896",
    "RequestId": "13f47dd0-1af9-4383-b401-dae18d6e99fc"
  }
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

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- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

QueryHunyuanTo3DProJob

Last updated: 2026-01-27 14:54:57

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

This API is used to intelligently generate 3D content based on the HunYuan Large Model and input text descriptions/images.

This API is used to provide 3 concurrent tasks by default, which can process 3 submitted tasks simultaneously. The next task can be processed only after the previous task is completed.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: QueryHunyuanTo3DProJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
JobId	Yes	String	Task ID.

3. Output Parameters

Parameter Name	Type	Description
Status	String	Task status: WAIT: waiting RUN: running; FAIL: failed DONE: successful
ErrorCode	String	Error code
ErrorMessage	String	Error message
ResultFile3Ds	Array of File3D	Describes the generated 3d file array
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Querying a 3D Task

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: QueryHunyuanTo3DProJob
<Common request parameters>

{
  "JobId": "1375367755519696896"
}
```

Output Example

```
{
  "Response": {
    "ErrorCode": "",
    "ErrorMessage": "",
    "RequestId": "e7c1aa24-7bcc-405e-9dce-c6047b9f6688",
    "ResultFile3Ds": [],
```

```
"Status": "RUN"  
}  
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

- [Tencent Cloud SDK 3.0 for Python](#)
- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

SubmitHunyuan3DPartJob

Last updated: 2026-01-22 10:59:59

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

This API is used to automatically perform component identification and generation based on the model structure after inputting a 3D model file.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: SubmitHunyuan3DPartJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
File	No	InputFile3D	Recommends inputting 3D models generated by AIGC Recommended file size not greater than 100MB Face count not greater than 30,000 Only supports FBX format

3. Output Parameters

Parameter Name	Type	Description
JobId	String	Task id (valid period: 24 hours)
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Submitting a Component Splitting Task Example

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: SubmitHunyuan3DPartJob
<Common request parameters>

{
  "File": {
    "Type": "FBX",
    "Url": "https://vcg-test-1258344699.cos.ap-guangzhou.myqcloud.com/test/3d/test.fbx"
  }
}
```

Output Example

```
{
  "Response": {
    "JobId": "1384898587778465792",
    "RequestId": "5265eb4a-0f4f-4cb1-9b3d-d9f1fb9347d2"
  }
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

- [Tencent Cloud SDK 3.0 for Python](#)
- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

QueryHunyuan3DPartJob

Last updated: 2026-01-22 11:45:25

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

This API is used to query the generation task of a component.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: QueryHunyuan3DPartJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
JobId	Yes	String	Task ID

3. Output Parameters

Parameter Name	Type	Description
----------------	------	-------------

Status	String	Task status: WAIT: waiting RUN: running FAIL: failed DONE: successful Example value: RUN
ErrorCode	String	Error code
ErrorMessage	String	Error message
ResultFile3Ds	Array of File3D	Generates the file URL with a valid period of 1 day
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Splitting the Task of Query Component

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: QueryHunyuan3DPartJob
<Common request parameters>

{
  "JobId": "1315932989749215232"
}
```

Output Example

```
{
  "Response": {
    "ErrorCode": "",
    "ErrorMessage": "",
    "RequestId": "5fec6e4d-9780-4372-815a-15490cb2ed5a",
    "ResultFile3Ds": [
      {
```

```
"Type": "OBJ",
"Url": "https://cos.ap-guangzhou.tencentcos.cn/xxx.obj"
},
{
  "Type": "FBX",
  "Url": "https://cos.ap-guangzhou.tencentcos.cn/xxx.fbx"
},
{
  "Type": "IMAGE",
  "Url": "https://cos.ap-guangzhou.tencentcos.cn/xxx.png"
}
],
"Status": "DONE"
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

- [Tencent Cloud SDK 3.0 for Python](#)
- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

Submit3DSmartTopologyJob

Last updated: 2026-01-26 15:29:55

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

The SmartTopology API uses the Polygen 1.5 model. After manually inputting a 3D high-poly model, it can generate a neat 3D model with lower polygon count.

1 concurrent is provided by default, which means 1 submitted task can be processed simultaneously. The next task can be processed only after the previous task is completed.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: Submit3DSmartTopologyJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
File3D	Yes	InputFile3D	Source 3D file model link Reference value:glb,obj One of the file formats is required. Url: file size cannot exceed 200MB.

			3D model requirements: complex models and topologized models do not support face reduction. recommended input is non-topologized high-poly models, such as those generated by hunyuan 3d. high applicability categories include hard surface, game character, prop, daily objects.
PolygonType	No	String	Polygon type, indicates the model surface is composed of grid components, defaults to triangle Reference value: triangle: triangle face quadrilateral: triangular and quadrilateral mixed face
FaceLevel	No	String	Reduction level bit type valid values: high, medium, low

3. Output Parameters

Parameter Name	Type	Description
JobId	String	Task ID.
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Calling the API

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: Submit3DSmartTopologyJob
<Common request parameters>

{
  "File3D": {
    "Url": "https://cos.ap-guangzhou.myqcloud.com/xxx.glb",
    "Type": "GLB"
```

```
}  
}
```

Output Example

```
{  
  "Response": {  
    "JobId": "1405140131170074624",  
    "RequestId": "f1dcd98c-47d2-47fe-81f2-26f4e0fec5d7"  
  }  
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

- [Tencent Cloud SDK 3.0 for Python](#)
- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

Describe3DSmartTopologyJob

Last updated: 2026-01-26 15:29:55

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

The SmartTopology API uses the Polygon 1.5 model. After manually inputting a 3D high-poly model, it can generate a neat 3D model with lower polygon count.

1 concurrent is provided by default, which means 1 submitted task can be processed simultaneously. The next task can be processed only after the previous task is completed.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: Describe3DSmartTopologyJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
JobId	Yes	String	Task ID.

3. Output Parameters

Parameter Name	Type	Description
Status	String	Task status. WAIT: waiting; RUN: running; FAIL: failed; DONE: successful. example value: RUN.
ErrorCode	String	Error code.
ErrorMessage	String	Error message.
ResultFile3Ds	Array of File3D	File generation URL address, with a valid period of 1 day.
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Calling the API

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: Describe3DSmartTopologyJob
<Common request parameters>

{
  "JobId": "1405140131170074624"
}
```

Output Example

```
{
  "Response": {
    "ErrorCode": "",
    "ErrorMessage": "",
    "ResultFile3Ds": [
      {
        "Type": "OBJ",
        "Url": "https://cos.ap-singapore.tencentcos.cn/xxx.obj"
      }
    ]
  }
}
```

```
}  
],  
"Status": "DONE",  
"RequestId": "75882497-fd9d-4b80-a2c0-1d7abf86e040"  
}  
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

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- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

SubmitHunyuanTo3DRapidJob

Last updated: 2026-02-13 16:53:10

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

This API is used to intelligently generate 3D content based on the HunYuan Large Model with input text descriptions or images.

This API is used to provide 1 concurrent task by default, which means only 1 submitted task can be processed simultaneously. The next task can be processed only after the previous task is completed.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: SubmitHunyuanTo3DRapidJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
Prompt	No	String	Text-To-3D, description of 3D content, forward Prompt content Supports up to 200 utf-8 characters

			Either ImageBase64, ImageUrl, or Prompt is required, and Prompt cannot coexist with ImageBase64/ImageUrl
ImageBase64	No	String	Input image Base64 data Size: unilateral resolution requirement not less than 128, not greater than 5000, size not greater than 6mb (after encoding, size increases by approximately 30%). format: jpg, png, jpeg, webp Imagebase64, imageurl, and Prompt are required, but Prompt and imagebase64/imageurl cannot coexist
ImageUrl	No	String	Input image Url size: Unilateral resolution requirement not less than 128, not greater than 5000. size not greater than 8mb Format: jpg, png, jpeg, webp Imagebase64, imageurl, and Prompt are required, and Prompt cannot coexist with imagebase64/imageurl
ResultFormat	No	String	Returns the 3D file format. valid values: OBJ, GLB, STL, USDZ, FBX, MP4, GIF Recommended input models below 50W, may timeout when selecting USDZ, MP4, or GIF format Example value: STL
EnablePBR	No	Boolean	Specifies whether PBR material generation is enabled, false by default.
EnableGeometry	No	Boolean	Specifies whether to enable the single geometry generation option When enabled, it generates a 3D model without textures (white model) When enabled, the generative model does not support OBJ format Default model file format is GLB

3. Output Parameters

Parameter Name	Type	Description
JobId	String	Task ID (valid period: 24 hours)
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request

will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Submitting a Task Call Example

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: SubmitHunyuanTo3DRapidJob
<Common request parameters>

{
  "ImageUrl": "https://***.cos.ap-guangzhou.myqcloud.com/***.jpg"
}
```

Output Example

```
{
  "Response": {
    "JobId": "1335141824121323520",
    "RequestId": "5f84f1df-7a29-4f93-8dc0-d9e6dea6bff9"
  }
}
```

5. Developer Resources

SDK

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- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)

- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

QueryHunyuanTo3DRapidJob

Last updated: 2026-02-13 16:53:11

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

This API is used to intelligently generate 3D content based on the HunYuan Large Model with input text descriptions or images.

This API is used to provide 1 concurrent task by default, which means only 1 submitted task can be processed simultaneously. The next task can be processed only after the previous task is completed.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: QueryHunyuanTo3DRapidJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
JobId	No	String	Task ID.

3. Output Parameters

Parameter Name	Type	Description
Status	String	Task status: WAIT: waiting RUN: running FAIL: failed DONE: successful
ErrorCode	String	Error code
ErrorMessage	String	Error message
ResultFile3Ds	Array of File3D	Specifies the generated 3D file array
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Query Task Sample Call

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: QueryHunyuanTo3DRapidJob
<Common request parameters>

{
  "JobId": "1336255233494892544"
}
```

Output Example

```
{
  "Response": {
    "ErrorCode": "",
    "ErrorMessage": "",
    "RequestId": "3020cd6c-4ad6-4df3-8560-c233f35d6221",
  }
}
```

```
"ResultFile3Ds": [  
  {  
    "PreviewImageUrl": "https://xxx.cos.ap-guangzhou.tencentcos.cn/xxx.png",  
    "Type": "OBJ",  
    "Url": "https://xxx.cos.ap-guangzhou.tencentcos.cn/xxx.zip"  
  },  
  {  
    "Status": "DONE"  
  }  
]
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

- [Tencent Cloud SDK 3.0 for Python](#)
- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

SubmitHunyuanTo3DTextureEditJob

Last updated: 2026-02-13 16:53:10

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

After inputting the 3D model, redraw the texture of the 3D model based on semantics or images, and output a redrawn texture image.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: SubmitHunyuanTo3DTextureEditJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
File3D	Yes	InputFile3D	File URL of the 3D model file that requires texture edit Supported formats: FBX 3D model limit: less than 100000 faces
Image	No	ImageInfo	Reference image for 3D model texture editing: Base64 data and image Url Either Base64 or Url must be provided. if both are provided,

			Url prevails Image restrictions: unilateral resolution less than 4096 and greater than 128, converted into Base64 string less than 10MB Format support: jpg, jpeg, png
Prompt	No	String	Describes texture editing, forward prompt content Supports up to 1024 utf-8 characters Either image or prompt is required, and prompt and image cannot coexist Example value: a kitten
EnablePBR	No	Boolean	Whether to enable the PBR texture parameter, only supports enabling when entering the Prompt Example value: true

3. Output Parameters

Parameter Name	Type	Description
JobId	String	Task id (valid period 24 hours)
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Submitting Texture Edit Task Example

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: SubmitHunyuanTo3DTextureEditJob
<Common request parameters>

{
  "File3D": {
    "Type": "FBX",
```

```
"Url": "https://vcg-test-1258344699.cos.ap-guangzhou.myqcloud.com/test/3d/test.fb  
x"  
}  
}
```

Output Example

```
{  
  "Response": {  
    "JobId": "1384898587778465792",  
    "RequestId": "5265eb4a-0f4f-4cb1-9b3d-d9f1fb9347d2"  
  }  
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

- [Tencent Cloud SDK 3.0 for Python](#)
- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

QueryHunyuanTo3DTextureEditJob

Last updated: 2026-02-13 16:53:11

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

Query 3D texture edit tasks, redraw the texture of the 3D model based on semantics or images, and output a redrawn texture image.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: QueryHunyuanTo3DTextureEditJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
JobId	Yes	String	Task ID

3. Output Parameters

Parameter Name	Type	Description
Status	String	Task status WAIT: waiting; RUN: running; FAIL: failed; DONE: successful; Example value: RUN
ErrorCode	String	Error code
ErrorMessage	String	Error message
ResultFile3Ds	Array of File3D	The URL address for generating texture files is valid for 1 day.
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Sample Code for Querying Texture Edit Tasks

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: QueryHunyuanTo3DTextureEditJob
<Common request parameters>

{
  "JobId": "1315932989749215232"
}
```

Output Example

```
{
  "Response": {
    "ErrorCode": "",
    "ErrorMessage": "",
```

```
"RequestId": "5fec6e4d-9780-4372-815a-15490cb2ed5a",
"ResultFile3Ds": [
  {
    "Type": "OBJ",
    "Url": "https://cos.ap-guangzhou.tencentcos.cn/xxx.obj"
  },
  {
    "Type": "FBX",
    "Url": "https://cos.ap-guangzhou.tencentcos.cn/xxx.fbx"
  },
  {
    "Type": "IMAGE",
    "Url": "https://cos.ap-guangzhou.tencentcos.cn/xxx.png"
  }
],
"Status": "DONE"
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

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- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

SubmitHunyuanTo3DUVJob

Last updated: 2026-02-03 20:41:45

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

After inputting the model, UV unfolding can be performed based on the model texture to output the corresponding UV texture

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: SubmitHunyuanTo3DUVJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
File	Yes	InputFile3D	File URL of the 3D model requiring UV unfold Supported formats: FBX, OBJ, GLB 3D model limit: less than 30000 faces

3. Output Parameters

Parameter Name	Type	Description
JobId	String	Task ID.
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Submitting a UV Unfolding Task Example

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: SubmitHunyuanTo3DUVJob
<Common request parameters>

{
  "File": {
    "Type": "GLB",
    "Url": "https://vcg-test-1258344699.cos.ap-guangzhou.myqcloud.com/test/3d/test.glb"
  }
}
```

Output Example

```
{
  "Response": {
    "JobId": "1384898587778465792",
    "RequestId": "5265eb4a-0f4f-4cb1-9b3d-d9f1fb9347d2"
  }
}
```

5. Developer Resources

SDK

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- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

DescribeHunyuanTo3DUVJob

Last updated: 2026-02-03 20:41:47

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

Query component splitting tasks.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: DescribeHunyuanTo3DUVJob.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
JobId	Yes	String	Task ID

3. Output Parameters

Parameter Name	Type	Description
----------------	------	-------------

Status	String	Task status WAIT: waiting; RUN: running; FAIL: failed; DONE: successful; Example value: RUN
ErrorCode	String	Error code
ErrorMessage	String	Error message
ResultFile3Ds	Array of File3D	File generation URL address, with a valid period of 1 day
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 Querying UV Display Task Example

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: DescribeHunyuanTo3DUVJob
<Common request parameters>

{
  "JobId": "1315932989749215232"
}
```

Output Example

```
{
  "Response": {
    "ErrorCode": "",
    "ErrorMessage": "",
    "RequestId": "5fec6e4d-9780-4372-815a-15490cb2ed5a",
    "ResultFile3Ds": [
      {
```

```
"Type": "OBJ",
"Url": "https://cos.ap-guangzhou.tencentcos.cn/xxx.obj"
},
{
  "Type": "FBX",
  "Url": "https://cos.ap-guangzhou.tencentcos.cn/xxx.fbx"
},
{
  "Type": "IMAGE",
  "Url": "https://cos.ap-guangzhou.tencentcos.cn/xxx.png"
}
],
"Status": "DONE"
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

- [Tencent Cloud SDK 3.0 for Python](#)
- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

Convert3DFormat

Last updated: 2026-02-13 16:53:12

1. API Description

Domain name for API request: hunyuan.intl.tencentcloudapi.com.

After inputting a 3D model file, the 3D model file format can be switched.

A maximum of 20 requests can be initiated per second for this API.

We recommend you to use API Explorer

[Try it](#)

API Explorer provides a range of capabilities, including online call, signature authentication, SDK code generation, and API quick search. It enables you to view the request, response, and auto-generated examples.

2. Input Parameters

The following request parameter list only provides API request parameters and some common parameters. For the complete common parameter list, see [Common Request Parameters](#).

Parameter Name	Required	Type	Description
Action	Yes	String	Common Params . The value used for this API: Convert3DFormat.
Version	Yes	String	Common Params . The value used for this API: 2023-09-01.
Region	Yes	String	Common Params . For more information, please see the list of regions supported by the product.
File3D	Yes	String	3D file url address. model file size not greater than 60 mb Supports fbx, obj, and glb format 3d file input
Format	Yes	String	Returns the 3D file format. valid values: STL, USDZ, FBX, MP4, GIF Recommended input models below 50W, may timeout when selecting USDZ, MP4, or GIF format Example value: STL.

3. Output Parameters

Parameter Name	Type	Description
ResultFile3D	String	3D file address
RequestId	String	The unique request ID, generated by the server, will be returned for every request (if the request fails to reach the server for other reasons, the request will not obtain a RequestId). RequestId is required for locating a problem.

4. Example

Example1 API Request Example

Input Example

```
POST / HTTP/1.1
Host: hunyuan.intl.tencentcloudapi.com
Content-Type: application/json
X-TC-Action: Convert3DFormat
<Common request parameters>

{
  "Format": "STL",
  "File3D": "https://cos.ap-guangzhou.tencentcos.cn/xxx.zip"
}
```

Output Example

```
{
  "Response": {
    "RequestId": "78722611-9e48-43fd-8e27-2672bae39d7b",
    "ResultFile3D": "https://cos.ap-guangzhou.tencentcos.cn/xxx.stl"
  }
}
```

5. Developer Resources

SDK

TencentCloud API 3.0 integrates SDKs that support various programming languages to make it easier for you to call APIs.

- [Tencent Cloud SDK 3.0 for Python](#)
- [Tencent Cloud SDK 3.0 for Java](#)
- [Tencent Cloud SDK 3.0 for PHP](#)
- [Tencent Cloud SDK 3.0 for Go](#)
- [Tencent Cloud SDK 3.0 for Node.js](#)
- [Tencent Cloud SDK 3.0 for .NET](#)
- [Tencent Cloud SDK 3.0 for C++](#)

Command Line Interface

- [Tencent Cloud CLI 3.0](#)

6. Error Code

There is no error code related to the API business logic. For other error codes, please see [Common Error Codes](#).

Data Types

Last updated: 2026-02-13 16:53:12

ErrorMsg

Runtime exception message.

Used by actions: ChatTranslations.

Name	Type	Description
Msg	String	Error message.
Code	Integer	Error code. 4000 internal service error. 4001 request model timeout.

File3D

3D model related files, files or textures.

Used by actions: Describe3DSmartTopologyJob, DescribeHunyuanTo3DUVJob, QueryHunyuan3DPartJob, QueryHunyuanTo3DProJob, QueryHunyuanTo3DRapidJob, QueryHunyuanTo3DTextureEditJob.

Name	Type	Required	Description
Type	String	No	3D model or texture. valid values: GIF or OBJ or Image.
Url	String	No	Specifies the file Url (valid for 24 hours).
PreviewImageUrl	String	No	Preview image Url.

ImageInfo

Image information.

Used by actions: SubmitHunyuanTo3DTextureEditJob.

Name	Type	Required	Description
------	------	----------	-------------

Url	String	No	Image Url.
Base64	String	No	Image Base64.

InputFile3D

3D file.

Used by actions: Submit3DSmartTopologyJob, SubmitHunyuan3DPartJob, SubmitHunyuanTo3DTextureEditJob, SubmitHunyuanTo3DUVJob.

Name	Type	Required	Description
Url	String	Yes	Specifies the file Url with a valid period of 24 hours.
Type	String	Yes	Specifies the file format.

PromptTokensDetails

Specifies the detailed information of the input token.

Used by actions: ChatTranslations.

Name	Type	Description
CachedTokens	String	The number of cache tokens.

Reference

Translate dialogue reference example.

Used by actions: ChatTranslations.

Name	Type	Required	Description
Type	String	No	Translate text type, enumerate "sentence" means sentence, "term" means terminology.
Text	String	No	Original.
Translation	String	No	Translation.

TranslationChoice

Translate the reply returned by the translation api, support multiple.

Used by actions: ChatTranslations.

Name	Type	Description
FinishReason	String	End flag, can be stop or sensitive. stop means output ends normally. sensitive only appears when streaming output review is enabled, indicating security review not passed.
Index	Integer	Index value, used when streaming.
Delta	TranslationDelta	Incremental return value used when streaming this field.
Message	TranslationMessage	Return value, used when non-streaming.

TranslationDelta

Translate the API response content (streaming return).

Used by actions: ChatTranslations.

Name	Type	Description
Role	String	Role name.
Content	String	Content details.

TranslationMessage

Translate the session content via translation api.

Used by actions: ChatTranslations.

Name	Type	Description
Role	String	Role. valid values: system, user, assistant, tool.
Content	String	Text content.

Usage

Token quantity.

Used by actions: ChatTranslations.

Name	Type	Description
PromptTokens	Integer	Input Token quantity.
CompletionTokens	Integer	Output Token quantity.
TotalTokens	Integer	Total Token quantity.
PromptTokensDetails	PromptTokensDetails	Details of the input token.

ViewImage

Multi-Perspective images.

Used by actions: SubmitHunyuanTo3DProJob.

Name	Type	Required	Description
ViewType	String	No	Specifies the viewing angle type. Valid values: back, left, right.
ViewImageUrl	String	No	Image Url.
ViewImageBase64	String	No	base64 address of the image.

Error Codes

Last updated: 2026-02-13 16:53:12

Feature Description

If there is an Error field in the response, it means that the API call failed. For example:

```
{
  "Response": {
    "Error": {
      "Code": "AuthFailure.SignatureFailure",
      "Message": "The provided credentials could not be validated. Please check your signature is correct."
    },
    "RequestId": "ed93f3cb-f35e-473f-b9f3-0d451b8b79c6"
  }
}
```

Code in Error indicates the error code, and Message indicates the specific information of the error.

Error Code List

Common Error Codes

Error Code	Description
ActionOffline	This API has been deprecated.
AuthFailure.InvalidAuthorization	<code>Authorization</code> in the request header is invalid.
AuthFailure.InvalidSecretId	Invalid key (not a TencentCloud API key type).
AuthFailure.MFAFailure	MFA failed.
AuthFailure.SecretIdNotFound	Key does not exist. Check if the key has been deleted or disabled in the console, and if not, check if the key is correctly entered. Note that whitespaces should not exist before or after the key.

AuthFailure.SignatureExpire	Signature expired. Timestamp and server time cannot differ by more than five minutes. Please ensure your current local time matches the standard time.
AuthFailure.SignatureFailure	Invalid signature. Signature calculation error. Please ensure you've followed the signature calculation process described in the Signature API documentation.
AuthFailure.TokenFailure	Token error.
AuthFailure.UnauthorizedOperation	The request is not authorized. For more information, see the CAM documentation.
DryRunOperation	DryRun Operation. It means that the request would have succeeded, but the DryRun parameter was used.
FailedOperation	Operation failed.
InternalError	Internal error.
InvalidAction	The API does not exist.
InvalidParameter	Incorrect parameter.
InvalidParameterValue	Invalid parameter value.
InvalidRequest	The multipart format of the request body is incorrect.
IpInBlacklist	Your IP is in uin IP blacklist.
IpNotInWhitelist	Your IP is not in uin IP whitelist.
LimitExceeded	Quota limit exceeded.
MissingParameter	A parameter is missing.
NoSuchProduct	The product does not exist.
NoSuchVersion	The API version does not exist.
RequestLimitExceeded	The number of requests exceeds the frequency limit.
RequestLimitExceeded.GlobalRegionUinLimitExceeded	Uin exceeds the frequency limit.

RequestLimitExceeded.IPLimitExceeded	The number of ip requests exceeds the frequency limit.
RequestLimitExceeded.UinLimitExceeded	The number of uin requests exceeds the frequency limit.
RequestSizeLimitExceeded	The request size exceeds the upper limit.
ResourceInUse	Resource is in use.
ResourceInsufficient	Insufficient resource.
ResourceNotFound	The resource does not exist.
ResourceUnavailable	Resource is unavailable.
ResponseSizeLimitExceeded	The response size exceeds the upper limit.
ServiceUnavailable	Service is unavailable now.
UnauthorizedOperation	Unauthorized operation.
UnknownParameter	Unknown parameter.
UnsupportedOperation	Unsupported operation.
UnsupportedProtocol	HTTP(S) request protocol error; only GET and POST requests are supported.
UnsupportedRegion	API does not support the requested region.

Service Error Codes

Error Code	Description
FailedOperation.EngineRequestTimeout	Request timeout at the engine layer. try again later.
FailedOperation.EngineServerError	Internal error at the engine layer. try again later.
FailedOperation.EngineServerLimitExceeded	Request exceeds the limit at the engine layer. try again later.
FailedOperation.FreeResourcePackExhausted	The remaining free resource package quota is exhausted. purchase a resource pack or enable post-payment.
FailedOperation.ResourcePackExhausted	The remaining resource package quota is exhausted. purchase a resource pack or enable post-payment.

FailedOperation.ServiceNotActivated	The service is not activated. go to the console to apply for a trial.
FailedOperation.ServiceStop	User proactively stopped services.
FailedOperation.ServiceStopArrears	Service suspended due to overdue payments.
InvalidParameterValue.Model	Model does not exist.